

Mycobacterial Metallophosphatase MmpE Acts as a Nucleomodulin to Regulate Host Gene Expression and Promote Intracellular Survival

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eLife Assessment

This manuscript is **useful** as it demonstrates that Rv2577, a $\text{Fe}^{3+}/\text{Zn}^{2+}$ -dependent metallophosphatase, is secreted by *Mycobacterium bovis* BCG and localizes to the nucleus of mammalian cells, altering transcriptional and inflammatory responses. However, the study is **incomplete** as it lacks activity validation in macrophage cells and with virulent *Mycobacterium tuberculosis* strains. It is necessary to confirm Rv2577 secretion from a virulent strain and to clarify the direct or indirect role of MmpE in modulating host pathways, together with mechanistic insight into how MmpE influences lysosomal biogenesis and trafficking.

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Abstract

Mycobacterium tuberculosis, the causative agent of tuberculosis, remains a major global health challenge. Nucleomodulins, bacterial effectors that target the host cell nuclei, are increasingly recognized as key virulence factors, but their roles in mycobacterial pathogenesis remain incompletely elucidated. Here, we characterize a hypothetical protein Rv2577 (designated MmpE) not only as a $\text{Fe}^{3+}/\text{Zn}^{2+}$ -dependent metallophosphatase but also as a critical nucleomodulin involved in immune evasion and intracellular persistence. MmpE utilizes two nuclear localization signals, RRR^{20–22} and RRK^{460–462}, to enter the host cell nucleus, where it binds to the promoter region of the vitamin D receptor (VDR) gene, thereby inhibiting host inflammatory gene expression. Additionally, MmpE represses the PI3K–Akt–mTOR signaling pathway, which arrests phagosome maturation. These actions collectively facilitate immune suppression and promote mycobacterial survival in macrophages and in mice. Our findings identify MmpE as a conserved nucleomodulin in mycobacteria and reveal a novel mechanism of MmpE-mediated intracellular survival.

Introduction

Mycobacterium tuberculosis (Mtb), the etiological agent of tuberculosis, has posed a persistent global health burden for centuries. Despite significant advances in elucidating the molecular mechanisms of its pathogenesis, current research has largely focused on effector proteins that modulate host cell membrane and cytoplasmic processes (Nisa *et al.*, 2022 [↗](#); Nasiri and Venketaraman, 2025 [↗](#)). In contrast, investigations into effector proteins that target the host cell nucleus, the central hub for gene regulation and cellular control, remain comparatively limited.

Nucleomodulins are a class of bacterial effector proteins that can translocate into the host nucleus, where they modulate nuclear processes to promote pathogen survival (Chai *et al.*, 2020 [↗](#); Bierre and Cossart, 2021 [↗](#)). These proteins affect host gene expression by directly interacting with chromatin or by mimicking transcription factors, chromatin modifiers, or other nuclear regulators (Li *et al.*, 2013 [↗](#); Rolando *et al.*, 2013 [↗](#); Mitra *et al.*, 2018 [↗](#); Hanford *et al.*, 2021 [↗](#)). For instance, LntA from *Listeria monocytogenes* directly binds the chromatin repressor BAHD1, disrupting its function and derepressing interferon-stimulated genes in a type III interferon-dependent manner, ultimately promoting bacterial persistence (Lebreton *et al.*, 2011 [↗](#); Lebreton *et al.*, 2014 [↗](#)). Similarly, Ank1 and Ank6 from *Orientia tsutsugamushi* localize to the host nucleus and facilitate the export of the NF- κ B p65 subunit via exportin 1, thereby suppressing the transcription of pro-inflammatory genes (Evans *et al.*, 2018 [↗](#); Steiert and Weber, 2025 [↗](#)). Although bacterial nucleomodulins utilize diverse strategies to manipulate host nuclear functions, the identity and functional characterization of *M. tuberculosis* nucleomodulins remain poorly defined.

Nucleomodulins employ various strategies to enter access to the host cell nucleus, including passive diffusion, hijacking host proteins that contain nuclear localization signals (NLSs), or using of their own NLSs to interact with the nuclear pore complex (Hanford *et al.*, 2021 [↗](#)). Among these mechanisms, NLSs play a particularly important role. Based on their amino acid composition, NLSs are broadly classified into classical (cNLS), non-classical (ncNLS), and other types (Lange *et al.*, 2008 [↗](#); Lu *et al.*, 2021 [↗](#)). Classical NLSs are further divided into monopartite (MP) and bipartite (BP) forms. Monopartite NLSs consist of a single cluster of 4–8 basic residues, primarily lysine (K) or arginine (R), with a typical consensus motif of K-K/R-X-K/R (Fontes *et al.*, 2000 [↗](#); Kosugi *et al.*, 2009 [↗](#); Cautain *et al.*, 2015 [↗](#)). Nuclear localization enables these effectors to manipulate host gene expression, disrupt immune signaling, and reprogram cellular processes to promote pathogen survival. For example, EspF from enteropathogenic *Escherichia coli* contains an N-terminal NLS that directs it to the nucleolus, where it interferes with ribosome biogenesis (Dean *et al.*, 2010 [↗](#); Singh *et al.*, 2018 [↗](#)). NUE, a SET domain-containing nucleomodulin secreted by *Chlamydia trachomatis*, also possesses an NLS that mediates its nuclear localization, where it functions as a histone lysine methyltransferase (HKMTase) targeting host histones H2B, H3, and H4 (Pennini *et al.*, 2010 [↗](#); Jermy, 2010 [↗](#)).

Beyond their nuclear localization capabilities, some nucleomodulins directly affect host immunity through intrinsic enzymatic activities (Agarwal *et al.*, 2012 [↗](#)). For example, the nucleomodulin PtpA promotes Mtb survival within macrophages by dephosphorylating host cytoplasmic proteins (Wang *et al.*, 2017 [↗](#)). NleC from *E. coli*, a Zn-dependent metalloprotease, suppresses host immune responses by cleaving the p65 subunit of NF- κ B in the cytoplasm, thereby inhibiting its nuclear translocation, and can enter the nucleus to reduce nuclear p65 levels, blocking NF- κ B-dependent transcription of pro-inflammatory genes (Baruch *et al.*, 2011 [↗](#); Hodgson *et al.*, 2015 [↗](#)). Among these enzymatically active nucleomodulins, purple acid phosphatases (PAPs) represent a distinct and functionally significant, yet underexplored, subclass within the metallophosphatase superfamily. PAPs are characterized by a conserved β - α - β - α - β fold and five signature motifs (DxG, GDXXY, GNH[D/E], VXXH, GHXH), which coordinate seven metal-ligating residues at a binuclear metal center (Rodriguez *et al.*, 2014 [↗](#); Bhadouria *et al.*, 2017 [↗](#); Bhadouria and Giri, 2022 [↗](#)). These

enzymes have been extensively studied in plants and mammals, where they are involved in key biological processes such as phosphorus metabolism and the generation of reactive oxygen species (Comba *et al.*, 2013 [↗](#); Feder *et al.*, 2020 [↗](#); Liu *et al.*, 2025 [↗](#)). In contrast, only a few bacterial PAPs have been identified, such as BcPAP from *Burkholderia cenocepacia*, and their functions remain largely unknown (Yeung *et al.*, 2009 [↗](#)). Notably, in *Mtb*, MmpE is the only identified PAP protein (Schenk *et al.*, 2000 [↗](#)), but its specific contribution to infection and pathogenesis remain elusive.

In this study, we characterized MmpE as a conserved nucleomodulin in mycobacteria and demonstrated its nuclear translocation mediated by two NLSs, RRR^{20–22} and RRK^{460–462}. Upon entering the nuclear, MmpE binds to the promoter region of the VDR gene, a key regulator of antimicrobial responses, resulting in transcriptional repression of pro-inflammatory genes. In addition, MmpE inhibits phagosome maturation by suppressing the expression of genes in the PI3K–Akt–mTOR signaling pathway. These actions collectively facilitate increased intracellular survival in both macrophage and mouse. Beyond its nuclear function, MmpE functions as a Fe³⁺/Zn²⁺ metallophosphatase, with enzymatic activity reliant on the conserved residues His³⁴⁸ and Asp³⁵⁹, and partially contributes to intracellular survival. Taken together, our findings identify MmpE as a bifunctional effector that integrates immune modulation and enzymatic catalysis, underscoring its role in mycobacterial pathogenesis and its potential as a therapeutic target.

Results

NLSs are required for the nuclear translocation of MmpE

In the prior study, we systematically identified some conserved nucleomodulins in mycobacterium. Among them, the hypothetical protein Rv2577 (MmpE) was identified as a candidate exhibiting strong nuclear localization in mammalian cells (Chen *et al.*, 2025 [↗](#)). Based on these findings, we transfected HEK293T cells with a C-terminal EGFP-tagged MmpE (MmpE-EGFP) and monitored its transcriptional level and nuclear localization at different time points. As a result, a time-dependent increase in the MmpE-EGFP content was observed in the host nucleus (Figure 1A [↗](#), upper panel; Figure S1A [↗](#)). Nuclear–cytoplasmic fractionation assays further confirmed these results, showing a progressive enrichment of MmpE in the nuclear compartment over the course of transfection (Figure 1A [↗](#), bottom panel). Collectively, these results indicate that MmpE possesses intrinsic nuclear translocation ability.

MmpE contains two putative NLSs, N-terminus (RRR^{20–22}, NLS1) and C-terminus (RRK^{460–462}, NLS2) (Figure 1B [↗](#)). To evaluate the structural contribution of these motifs, we utilized AlphaFold to model the impact of NLS truncations on the overall conformation of MmpE. The predicted structures indicated that deletion of either NLS1, NLS2, or both did not obviously alter the protein fold (Figure S1B [↗](#)). Based on this analysis, we generated a series of NLS-deleted mutants, including MmpE^{ΔNLS1}, MmpE^{ΔNLS2}, and MmpE^{ΔNLS1–2}, to assess the functional relevance of each motif in nuclear localization. Subcellular localization of wild-type MmpE-EGFP and NLS deletion mutants was examined in HEK293T cells by fluorescence microscopy at 36 hpt (hours post-transfection). Deletion of either NLS1 or NLS2 alone significantly reduced the nuclear accumulation of MmpE-EGFP, with the effect of NLS2 deletion being more pronounced than that of NLS1. Simultaneous deletion of both NLS motifs (MmpE^{ΔNLS1–2}) completely abolished nuclear localization, resulting in exclusive cytoplasmic distribution (Figure 1C [↗](#) and D [↗](#)). These findings were further validated by immunoblotting of nuclear and cytoplasmic fractions. Under the similar transfection conditions, wild-type MmpE-EGFP was detected in both compartments, whereas MmpE^{ΔNLS1} and MmpE^{ΔNLS2} exhibited substantially reduced nuclear abundance. By contrast, MmpE^{ΔNLS1–2} was exclusively localized to the cytoplasmic fraction, suggesting that both NLS

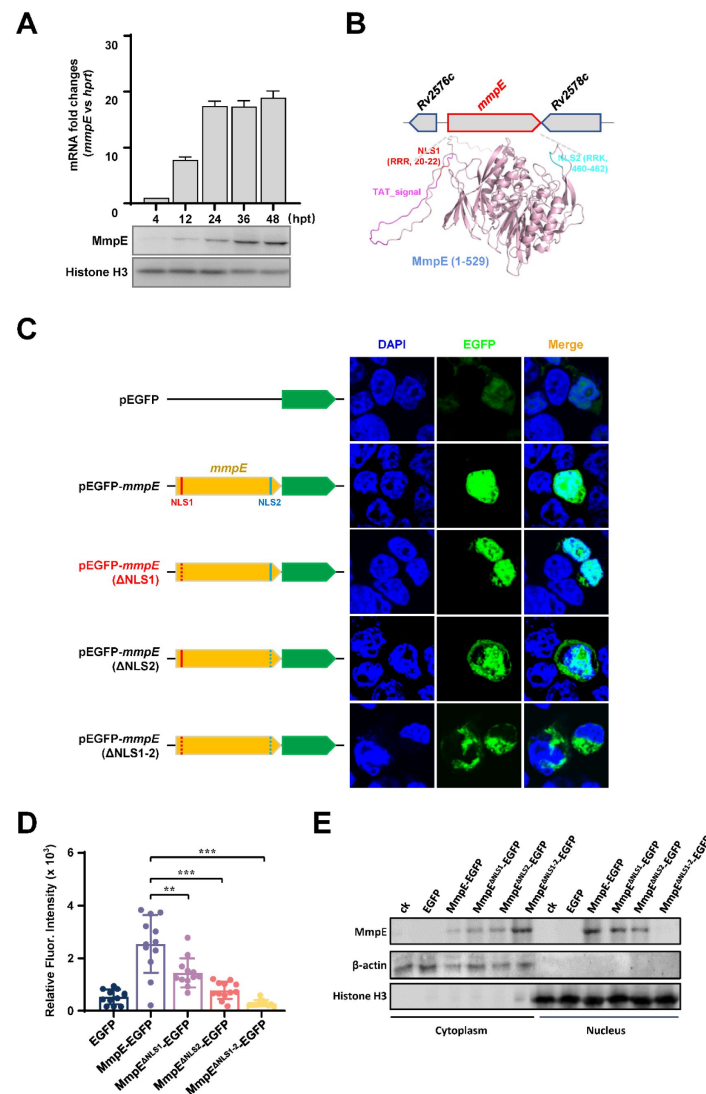


Figure 1.

NLSs are required for the nuclear translocation of MmpE.

(A) Expression and nuclear localization of *MmpE*. (upper) qRT-PCR analysis of *mmpE* mRNA in HEK293T cells over 48 h post-transfection. (bottom) Western blot analysis of nuclear fractions showing time-dependent nuclear accumulation of MmpE-EGFP. Histone H3 and β -actin served as nuclear and cytoplasmic markers, respectively.

(B) Domain architecture of MmpE. The schematic representation of MmpE shows its annotated functional domains, including a Tat signal peptide (1-54 amino acids, with a twin-arginine translocation motif) and two nuclear localization signals (NLS1: 20-22 amino acids; NLS2: 460-462 amino acids). The structure of the MmpE protein was predicted using AlphaFold 2.2.0, where the position of NLS1 is highlighted in red and NLS2 is highlighted in green.

(C) Subcellular localization of EGFP-tagged wild-type and NLS-deleted MmpE. (left) Schematic representation of EGFP-tagged constructs, including wild-type MmpE and its mutants (MmpE Δ NLS1, MmpE Δ NLS2, and MmpE Δ NLS1-2). (right) Confocal microscopy images of HEK293T cells transfected with the indicated constructs for 36 hpt. EGFP fluorescence (green) and nuclear staining with DAPI (blue) were visualized using an FLUOVIEW software (v5.0). Scale bar, 10 μ m. Images were acquired with a $\times 100$ oil immersion objective (NA = 1.4).

(D) Nuclear EGFP intensity of wild-type and mutant constructs in **(C)**. Data are shown as mean $\pm$ SD (n = 12 cells).

(E) Western blot analysis of nuclear and cytoplasmic fractions from HEK293T cells transfected with wild-type and mutant MmpE-EGFP confirmed their nuclear localization. MmpE-EGFP was detected using an anti-GFP antibody, and histone H3 and β -actin served as nuclear and cytoplasmic markers, respectively.

All experiments were performed in three biologically independent replicates. Statistical significance was assessed by two-tailed unpaired Student's *t*-tests, with $*P < 0.05$ considered statistically significant.

motifs are required for efficient nuclear import (**Figure 1E**). Taken together, our findings identify MmpE as a nucleomodulin, with nuclear import mediated by the N- and C-terminal NLS motifs.

Deletion of NLSs does not alter the phosphatase activity of MmpE

Bioinformatic analyses identified MmpE as a $\text{Fe}^{3+}/\text{Zn}^{2+}$ -metallophosphatase containing a purple acid phosphatase (PAP) motif (**Figure 2A**; **Figure S2A**). Multiple sequence alignment and phylogenetic reconstruction revealed that MmpE is highly conserved among mycobacterial species and harbors canonical motifs characteristic of metallophosphatases, including GD, GNHX, and GHXH (**Figure 2B**, **Figure S2B**). These motifs are typically associated with enzymes that catalyze phosphate hydrolysis via a binuclear metal ion-dependent mechanism, wherein residues such as Asp, His, and Asn coordinate essential metal cofactors, commonly Fe^{3+} , Mn^{2+} , or Zn^{2+} , at the active site.

To identify key amino acid residues of MmpE involved in metal coordination, AlphaFold was used to predict the structure of MmpE, followed by the molecular docking using PyMOL. As shown in **Figure 2C**, H^{348} and N^{359} were identified as candidate residues likely involved in metal ion coordination within the predicted active site. Subsequently, recombinant MBP-tagged MmpE deleting the Tat signal peptide ($\text{MmpE}^{\Delta\text{Tat}}$) was subjected to *in vitro* phosphatase assays using *p*-nitrophenyl phosphate (*p*-NPP) as the substrate. Reactions were supplemented with varying concentrations of Fe^{3+} and Zn^{2+} to evaluate their individual effects on MmpE enzymatic activity. The results demonstrated that, at equivalent concentrations, Fe^{3+} significantly enhanced MmpE activity compared to Zn^{2+} , with maximal enzymatic activity was observed at 50 μM Fe^{3+} (**Figure 2D**). Next, a double mutant ($\text{MmpE}^{\Delta\text{Tat-H}^{348}\text{N}^{359}\text{A}}$) was generated via site-directed mutagenesis to validate the functional importance of these residues. As shown in **Figure 2E**, the addition of 50 μM Fe^{3+} significantly increased the *in vitro* phosphatase activity of $\text{MmpE}^{\Delta\text{Tat}}$ compared to reactions without metal ions. However, the enzymatic activity of the $\text{MmpE}^{\Delta\text{Tat-H}^{348}\text{N}^{359}\text{A}}$ in the presence of 50 μM Fe^{3+} was markedly reduced and nearly indistinguishable from that of $\text{MmpE}^{\Delta\text{Tat}}$ under metal-free conditions. These results confirm that residues H^{348} and N^{359} are essential for Fe^{3+} binding and catalytic activity.

Given that MmpE relies on NLSs to enter the host nucleus, we examined whether deletion of NLS affects its phosphatase activity. Compared to wild-type MmpE, the NLS-deleted mutant showed no significant change in enzymatic activity (**Figure 2F**), which is consistent with the structural comparison results (**Figure S1B**). Collectively, these results suggest that MmpE is a highly conserved $\text{Fe}^{3+}/\text{Zn}^{2+}$ -dependent metallophosphatase in mycobacteria, and its enzymatic activity is independent of NLS.

The nuclear translocation and phosphatase activity of MmpE are essential for *M. bovis* BCG survival in macrophage cells

To investigate the role of MmpE during infection, we first generated recombinant *M. bovis* BCG strains expressing C-terminally Flag-tagged Ag85B (positive control) and MmpE. Immunoblot analysis confirmed high levels of Ag85B-2×Flag and MmpE-2×Flag in bacterial lysates, and both proteins were also detected in the culture supernatants (**Figure S3A**), indicating that MmpE is indeed a secreted protein.

Furthermore, a series of recombinant BCG strains were constructed, including wild-type (WT), *mmpE*-deleted mutant (ΔMmpE) (**Figure S3B**), ΔMmpE complemented with wild-type *mmpE* (Comp-MmpE), and ΔMmpE complemented with NLS-deleted mutant (Comp-MmpE $^{\Delta\text{NLS1}}$, Comp-MmpE $^{\Delta\text{NLS2}}$, and Comp-MmpE $^{\Delta\text{NLS1-2}}$). Deletion of NLS1 may reduce or abolish MmpE secretion (Stanley *et al.*, 2000; Buchanan *et al.*, 2001), while deletion of NLS2 impaired its nuclear localization. All strains exhibited similar growth rate under the normal culture conditions (**Figure S3C**). Subsequently, these strains were used to infect THP-1 macrophages. As shown in **Figure**

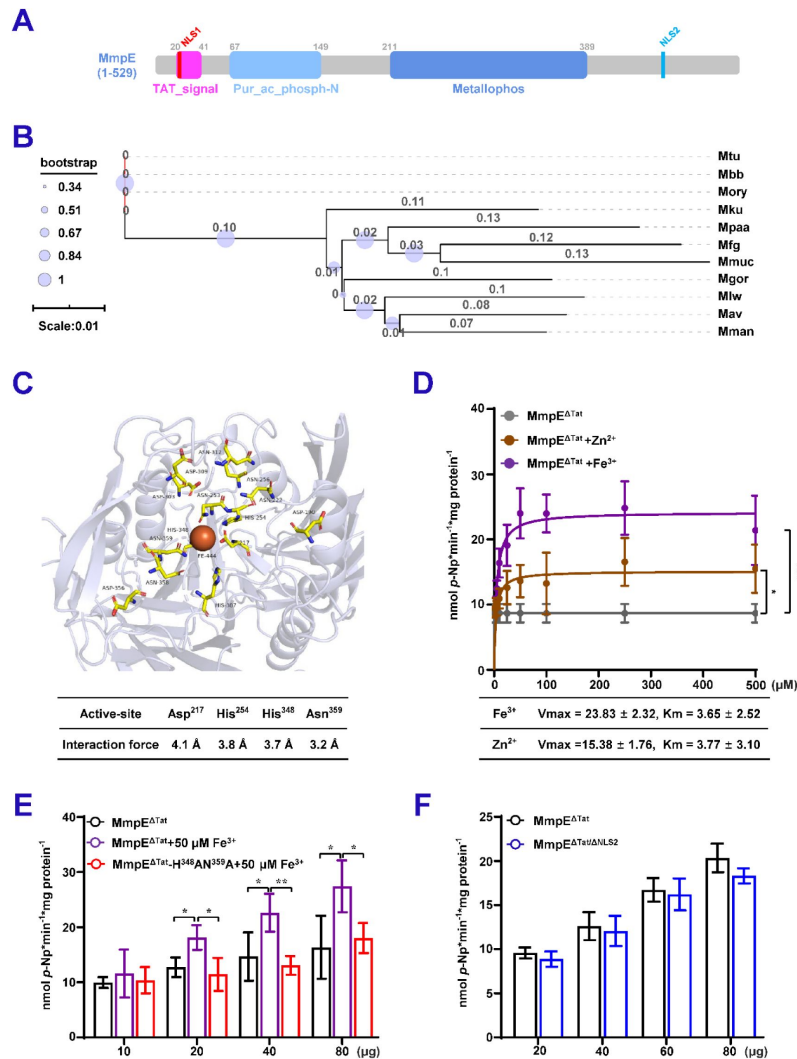


Figure 2.

Deletion of NLSs does not alter the phosphatase activity of MmpE.

(A) Domain architecture of MmpE. Schematic representation of MmpE with the annotated functional domains, including a Tat signal peptide (1–41 aa, twin-arginine translocation motif), a purple acid phosphatase domain (68–149 aa), a calcineurin-like phosphoesterase domain (211–389 aa) and two nuclear localization signals (NLS1: 20–22 aa; NLS2: 460–462 aa).

(B) Phylogenetic and structural conservation of MmpE. Neighbor-joining phylogenetic tree of MmpE homologs across Mycobacterium species (1,000 bootstrap replicates; values ≥50% shown). Species abbreviations: *M. tuberculosis* (Mtu), *M. bovis* BCG (Mbb), *M. orygis* (Mory), *M. kansasii* (Mku), *M. paratuberculosis* (Mpaa), *M. farcinogenes* (Mfg), *M. mucogenicum* (Mmuc), *M. vaccae* (Mvg), *M. lentiflavum* (Mlw), *M. avium* (Mav), *M. goodii* (Mgo), *M. neoaurum* (Mna), *M. indicus pranii* (Mip), *M. mageritensis* (Mma), *M. goodii* (Mgo), *M. neoaurum* (Mna), *M. indicus pranii* (Mip), *M. mageritensis* (Mma).

(C) Prediction of metal ion-binding residues in MmpE. Structural modeling and visualization of MmpE were performed using PyMOL. Conserved residues located in the putative metal binding pocket are shown as sticks and colored according to atom type. The predicted metal coordination site is highlighted, and key residues potentially involved in metal ion binding are labeled. Surface representation is included to illustrate the spatial accessibility of the binding pocket.

(D–F) Phosphatase activity assays of MmpE^{ΔTat} under various conditions. Phosphatase activity of MmpE^{ΔTat} (lacking the N-terminal Tat signal peptide) was measured using *p*-nitrophenyl phosphate (*p*-NPP) as the substrate in the presence of increasing concentrations (0–500 μM) of Fe³⁺ and Zn²⁺ (**D**). Phosphatase activity of increasing concentrations of MmpE^{ΔTat} and the mutant MmpE^{ΔTat-H³⁴⁸AN³⁵⁹A} was measured in the absence or presence of 50 μM Fe³⁺ (**E**). Phosphatase activity of increasing concentrations of MmpE^{ΔTat} and the double-NLS deletion mutant MmpE^{ΔTat/ΔNLS1-2} was measured under standard conditions (**F**).

Data represent mean ± SD of three biologically independent experiments, analyzed using two-tailed unpaired Student's *t*-tests (**P* < 0.05, ***P* < 0.01, ****P* < 0.001).

3A and **B**, the bacterial load of the Δ MmpE strain exhibited significantly reduced relative to the WT strain, and bacterial clearance was accelerated over time. The phenotype of Δ MmpE was rescued by complementation with wild-type gene *mmpE*. Meanwhile, the load of the Comp-MmpE^{ΔNLS2} strain displayed significantly lower compared to Comp-MmpE in both THP-1 and RAW264.7 macrophages, indicating that nuclear localization mediated by NLS2 is critical for MmpE function. Notably, the Comp-MmpE^{ΔNLS1} and Comp-MmpE^{ΔNLS1-2} strains demonstrated obvious higher load than those of the Δ MmpE strain, yet lower than those of the Comp-MmpE strain (**Figure 3C** and **D**). These results suggest that MmpE contributes to virulence not only through nuclear modulation but also through additional mechanisms.

Next, we investigated the role of the Fe³⁺/Zn²⁺-dependent metallophosphatase of MmpE on intracellular survival. As shown in **Figure S3D**, a double mutant strain (Comp-MmpE-H³⁴⁸AN³⁵⁹A) was constructed and used to infect macrophage cells. The results showed that this mutant exhibited significantly reduced intracellular load compared with the Comp-MmpE strain (**Figure 3E**), indicating that the phosphatase activity also plays an essential role in MmpE-mediated bacterial survival. To reveal the underlying mechanism, cytokine expression assays were conducted in infected macrophages. Compared with the WT and Comp-MmpE strains, cells infected with the Δ MmpE and Comp-MmpE^{ΔNLS1-2} strains showed markedly elevated levels of pro-inflammatory cytokines, including *IL1A*, *IL1B*, and *IL6* (**Figure 3F**).

Collectively, these results suggest that MmpE functions as a bifunctional effector, possessing both nuclear import capability and phosphatase activity, promoting immune evasion and enhancing intracellular survival during infection.

MmpE modulates host transcription network involved in inflammation response and lysosomal maturation

To investigate the role of the nucleomodulin MmpE on host gene expression, we performed RNA-seq assays on THP-1 macrophages infected with either wild-type (BCG/wt) or MmpE-deleted (*KommpE*) strains at 20 hpi (hours post-infection). As a result, a total of 175 differentially expressed genes (DEGs) were identified, with 142 genes upregulated and 33 genes downregulated in *KommpE*-infected macrophages compared to those infected with BCG/wt (**Figure 4A**, Table S1). These DEGs were significantly enriched in immune-related biological processes, particularly those associated with apoptosis and cell death, as well as molecular functions related to immune activation and cellular stress responses, such as pyrophosphatase activity and nucleoside-triphosphatase activity (**Figure 4B**). Further analysis using the Kyoto Encyclopedia of Genes and Genomes (KEGG) revealed that the DEGs were primarily involved in immune and inflammatory signaling pathways, including cytokine–cytokine receptor interaction, TNF signaling, Toll-like receptor signaling, chemokine signaling, NF-κB signaling, JAK–STAT signaling, and the PI3K–Akt signaling pathway. Additionally, several DEGs were enriched in pathways related to cell survival and apoptosis. Another group of DEGs was associated with host defense mechanisms, as evidenced by their enrichment in the lysosome pathway (**Figure 4C**). Together, these data highlight MmpE as a critical modulator of host transcriptional responses that facilitate mycobacterial immune evasion and intracellular survival.

To gain further insight into the functional relevance of the 175 differentially expressed genes, we performed protein–protein interaction (PPI) network analysis, focusing on protein-coding genes, which accounted for 38% of the total (**Figure S4A**). A tightly connected network centered around key immune regulatory, including IL23A, IL12B, CSF2, CD69, IDO1, and CEACAM1, which are critically involved in inflammatory cytokine production (**Figure 4D**). Consistently, these genes expression was upregulation in *KommpE*-infected macrophages (**Figure 4E**). Given that inflammatory signaling and lysosomal function are both essential for macrophage-mediated host defense against mycobacterial infection, we further examined the expression of representative inflammatory cytokines and key lysosomal maturation markers under the same experimental

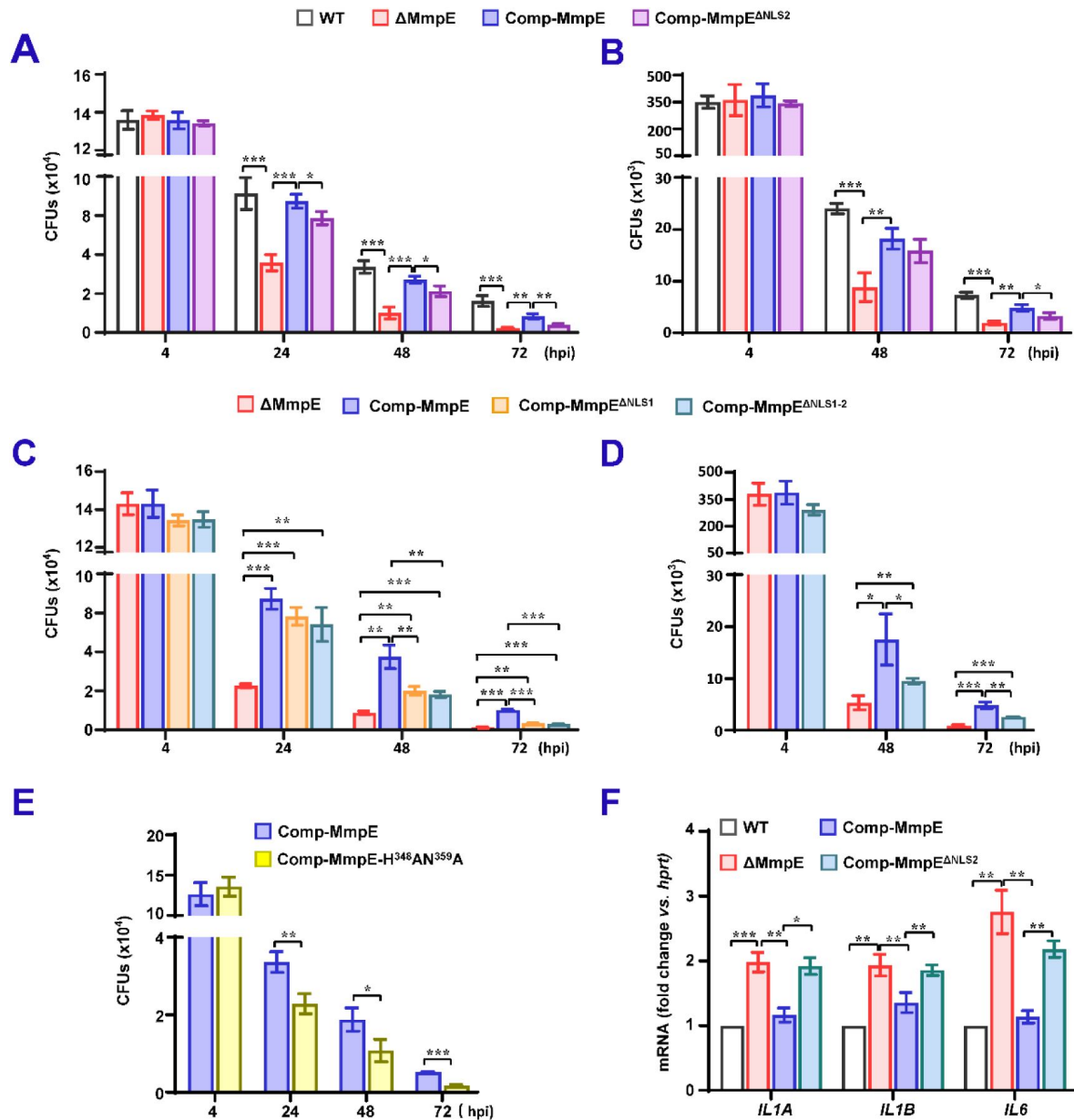


Figure 3.

The nuclear translocation and phosphatase activity of MmpE are essential for *M. bovis* BCG survival in macrophage cells.

(A–B) Intracellular survival of *M. bovis* BCG strains in human THP-1 macrophages (A) and murine RAW264.7 macrophages (B). Strains include wild-type BCG (WT), MmpE deletion mutant (ΔMmpE), wild-type MmpE complemented strain (Comp-MmpE), and NLS2-deleted complement strain (Comp-MmpE^{ΔNLS2}).

(C–D) Intracellular survival of *M. bovis* BCG strains in THP-1 (C) and RAW264.7 (D) macrophages. Strains include ΔMmpE, Comp-MmpE, and NLS-deficient complement strains (Comp-MmpE^{ΔNLS1} and Comp-MmpE^{ΔNLS1-2}).

(E) Intracellular survival of metal ion binding site mutants in THP-1 cells. Strains include Comp-MmpE and a phosphatase-deficient mutant (Comp-MmpE-H³⁴⁸AN³⁵⁹H).

(F) Inflammatory cytokine expression in infected THP-1 cells. mRNA levels of *IL-1α*, *IL-1β*, and *IL-6* were quantified by qRT-PCR 20 hpi with the indicated BCG strains. Data represent mean ± SD from three independent biological replicates. Statistical analysis was performed using two-tailed unpaired Student's *t*-tests (**P* < 0.05, ***P* < 0.01, ****P* < 0.001).

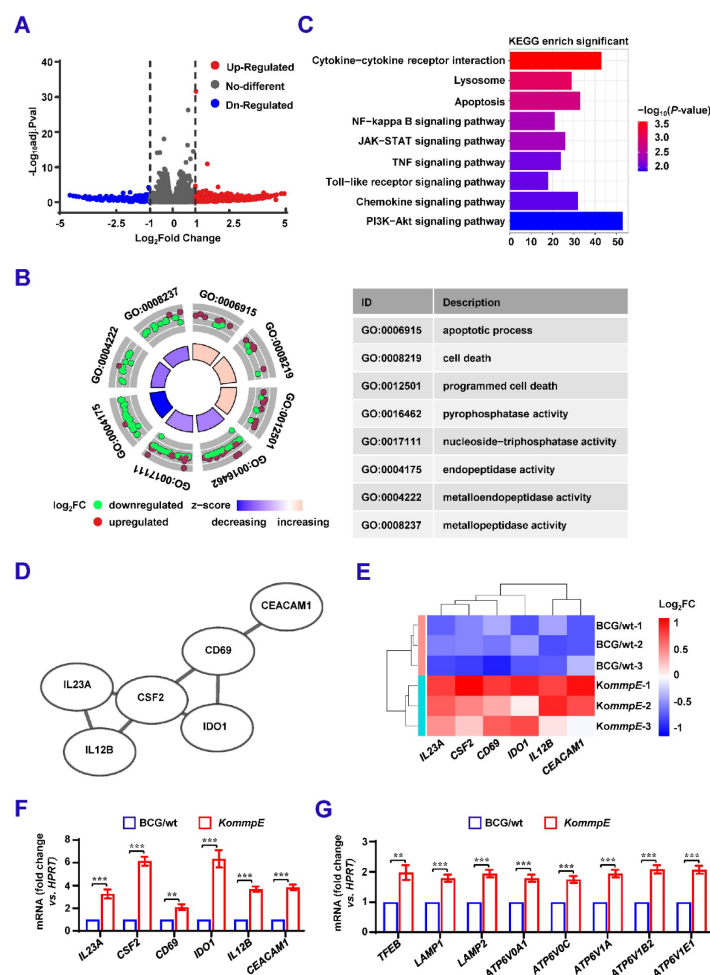


Figure 4.

MmpE modulates host transcription network involved in inflammation response and lysosomal maturation.

(A) Volcano plot of differentially expressed genes (DEGs). DEGs in *MmpE* deleted strain (*KommpE*) compared to wild-type BCG (BCG/wt) were visualized in a volcano plot. DEGs were defined as genes with $|\log_2(\text{fold change})| \geq 1$ and $P < 0.05$. The x-axis represents $\log_2(\text{fold change})$, and the y-axis shows $-\log_{10}(P\text{-value})$. Genes with significant upregulation and downregulation are highlighted.

(B) GO enrichment analysis of DEGs in *KommpE*-infected THP-1 cells versus BCG/wt. The circular plot shows enriched biological process and molecular function terms. Outer rings represent GO terms, with red and green dots indicating upregulated and downregulated DEGs, respectively. The inner ring color reflects z-scores (blue: low; pink: high). The table lists selected GO terms with their IDs and descriptions.

(C) KEGG enrichment analysis of DEGs in *KommpE*-infected THP-1 cells versus BCG/wt. The bar plot displays selected significantly enriched pathways. Bar length represents the number of DEGs associated with each pathway, and color indicates statistical significance based on $-\log_{10}(P\text{-value})$.

(D) Interaction network of DEGs related to immune regulation. Network construction was performed using STRING v12.0 and visualized with Cytoscape.

(E) Heatmap of immune-related DEGs in *KommpE*-infected and BCG/wt-infected THP-1 cells. $\log_2$ fold change values are shown across three biological replicates. Red and blue indicate upregulation and downregulation, respectively.

(F-G) Quantitative RT-PCR analysis of gene expression in infected THP-1 cells. THP-1 cells were infected with BCG/wt or *KommpE* strains for 20 hpi. mRNA levels of cytokine genes (F); mRNA levels of genes involved in lysosomal acidification and biogenesis (G).

Data are presented as mean $\pm$ SD of three biologically independent experiments, analyzed using two-tailed unpaired Student's *t*-tests (* $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$).

conditions, such as *TFEB*, *LAMP1*, *LAMP2*, and several V-ATPase subunit genes (*ATP6V0A1*, *ATP6V0C*, *ATP6V1A*, *ATP6V1B2*, and *ATP6V1E1*). Consistent with transcriptomic findings, the *KommpE* strain induced significantly higher expression of both inflammatory cytokines and lysosomal maturation-related genes compared to the WT strain (**Figure 4F** and **G**). Both the Comp-MmpE^{ANLS2} and Comp-MmpE^{ANLS1-2} strains showed significantly higher expression of these cytokine genes compared to the Comp-MmpE strain (**Figure S4B** and **C**). Taken together, these results suggest that MmpE manipulates host transcriptional networks governing inflammation and lysosomal function, thereby subverting antimicrobial immunity.

MmpE suppresses the expression of *VDR* and inhibits activation of the PI3K–Akt–mTOR signaling pathway

To elucidate the mechanism by which MmpE modulates host gene expression, we performed ChIP-seq in HEK293T cells transfected with EGFP-MmpE. 2,903 potential MmpE-binding sites (Table S2) were identified, with 18.49% (537) in intergenic regions and 39.44% (1145) in intragenic regions (**Figure 5A**). Chromosomal distribution revealed significant enrichment on chromosomes 1 and 2 (**Figure 5B**). Notably, 99.7% (2894) of these binding sites were associated with protein-coding regions. Among these, 1013 sites were located within 3 kb of transcription start sites (TSS), including promoter (≤ 1 kb, 63.18%), promoter (1–2 kb, 22.9%), and promoter (2–3 kb, 13.92%) (**Figure 5C** and **D**), suggesting that MmpE primarily targets promoter regions to regulate transcription. PPI analysis of the 1013 potential target genes revealed significant enrichment in protein kinase-related genes, such as *PRKCB*, *PLCG2* and *PIK3CB*, involved in regulating immune activation and inflammatory responses via the PI3K–Akt and mTOR signaling pathways, suggesting that MmpE may influence cell immune responses through a coordinated network of signaling pathways (**Figure S5A** and **B**). In parallel, we examined the expression of key components in this signaling axis in THP-1 macrophages infected with either BCG/wt or the *KommpE* strains. Notably, *KommpE* infection led to a significant upregulation of genes associated with the pathway, including *PIK3CA*, *PIK3CB*, *PIK3R1*, *mTOR*, *AKT1–3*, and *FOXO1/3* (**Figure S5C**). These genes encode critical regulators of cell survival and immune activation within the PI3K–Akt–mTOR axis.

To identify the direct target gene of MmpE in macrophages, we focused on analysis of the promoter regions (≤ 1 kb upstream of the transcription start site) of significantly upregulated genes with $|\text{Log}_2\text{FC}| > 1$ and $P < 0.05$. Four transcription factors were identified, including CREBBP, *VDR*, GREB1, and FOXP4 (**Figure 5E**). Integrative analysis using HOMER *de novo* motif prediction revealed that only the promoter region of *VDR*, a key anti-inflammatory gene (Aggeletopoulou *et al.*, 2022), harbored a potential MmpE-binding motif (**Figure 5F**). These findings suggest that MmpE may directly bind to the *VDR* promoter. To validate the hypothesis, we examined *VDR* expression in macrophages infected with recombinant strains. Compared to the BCG/wt, *VDR* expression was significantly downregulated in THP-1 cells infected with the *KommpE* strain (**Figure S5D**). To further test whether MmpE directly binds the *VDR* promoter, we cloned the *VDR* promoter region (–100 bp to +50 bp) and performed chromatin immunoprecipitation (ChIP) assays in HEK293T cells. While both *VDR* and *GAPDH* (control) sequences were present in input samples, *VDR* promoter DNA was detected exclusively in ChIP samples from EGFP-MmpE transfected cells (**Figure 5G** and **H**). Furthermore, the DNA-binding capacity of MmpE was unaffected by its phosphatase activity (**Figure S5E** and **F**). To confirm these results, we performed electrophoretic mobility shift assays (EMSA) using the same *VDR* promoter fragment. As shown in **Figure 5I** and **5J**, the shift bands corresponding to MmpE and *VDR* promoter complex were observed, and a corresponding increase was observed in the percentage of protein/DNA complexes. Collectively, these results provide strong evidence that MmpE directly interacts with the *VDR* promoter both *in vivo* and *in vitro* to repress host inflammatory signaling pathways and facilitating immune evasion during mycobacterial infection.

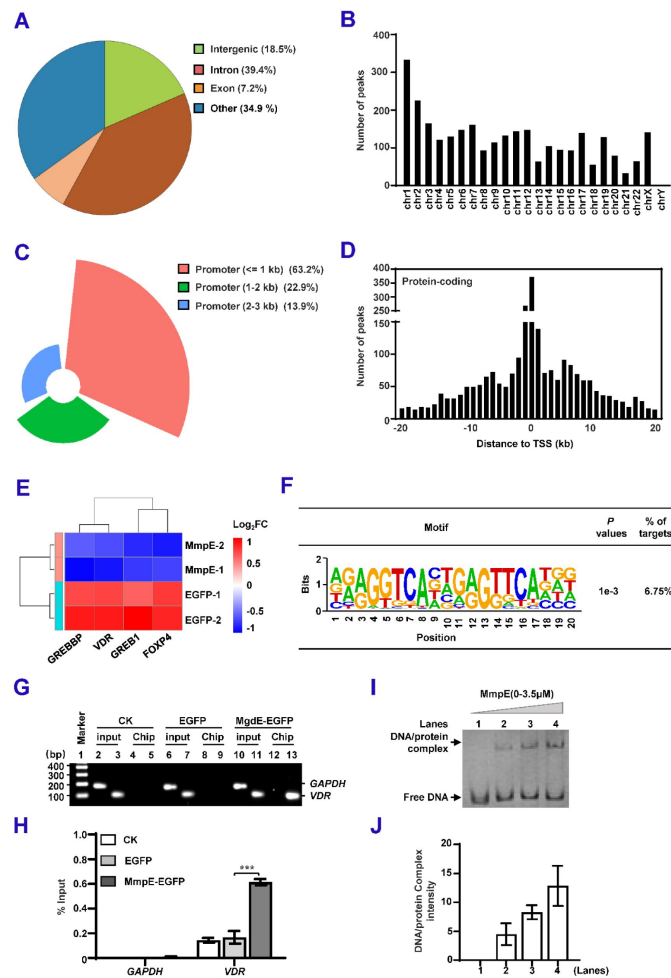


Figure 5.

MmpE suppresses the expression of *VDR* and inhibits activation of the PI3K–Akt–mTOR signaling pathway.

(A) Genomic distribution of potential MmpE-binding regions in HEK293T cells.
 (B) Distribution of the chromosomal location of MmpE-specific ChIP-seq signals (fold-enrichment > 9).
 (C) Biotype distribution of potential MmpE-binding regions in HEK293T cells.
 (D) The number and position of the potential MmpE-binding sites along 20 kb from the nearest TSS of the protein-coding genes.
 (E) Heatmap of transcription factors identified by MmpE ChIP-seq. Colors represent expression changes, with red indicating upregulation and blue indicating downregulation.
 (F) MmpE-binding motifs identified by *de novo* motif analysis of ChIP-seq peak sequences using HOMER. Motif enrichment *P*-values were calculated using TOMTOM. “% of targets” indicates the proportion of MmpE-bound regions containing each motif. Letter height reflects the frequency of each nucleotide at the corresponding position in the consensus motif.
 (G–H) ChIP-PCR and qPCR analyses of MmpE binding to the *VDR* promoter region. Chromatin was immunoprecipitated from HEK293T cells transfected with control vector (CK), EGFP, or MmpE–EGFP constructs. PCR was performed using primers targeting the *GAPDH* region (negative control) and the *VDR* promoter region, and products were analyzed by agarose gel electrophoresis (G). ChIP enrichment for *VDR* and *GAPDH* was quantified in HEK293T cells using qPCR, with enrichment calculated by the $2^{-\Delta\Delta C_t}$ method (H).
 (I–J) EMSA analysis showing the binding of MmpE to the human *VDR* promoter. The formation of DNA–protein complexes was observed with increasing concentrations of MmpE protein. Arrows indicate the positions of free DNA and DNA–protein complexes on the native polyacrylamide gel (I). Quantitative analysis of DNA–protein complexes in each lane (I) was performed by densitometry (J).
 Data are presented as mean $\pm$ SD of three biologically independent experiments. Statistical significance was determined by a two-tailed unpaired Student’s *t*-test ($***P < 0.001$).

To further characterize the transcriptional programs regulated by MmpE, we performed an integrative analysis of CUT&Tag and RNA-seq datasets. This analysis identified 298 DEGs with direct or proximal MmpE binding sites (Table S3; **Figure S6A** and **B**). Notably, genes such as *RHEX*, *PLXDC1*, and *GAS6* exhibited significant expression changes and were enriched in pathways associated with cell differentiation and small GTPase-mediated signaling (**Figure S6C** and **D**). These pathways play critical roles in shaping immune cell function, trafficking, and inflammatory responses during host–pathogen interactions (Li *et al.*, 2023; Ma *et al.*, 2024). Taken together, these data establish MmpE as a nucleomodulin that directly alters host transcriptional networks involved in immune regulation and cellular defense, thereby supporting mycobacterial survival within host cells.

Nuclear translocation of MmpE is essential for mycobacterial survival in mice

Next, C57BL/6 mice were intranasally infected with *M. bovis* BCG strains, including wild-type (WT), Δ MmpE, Comp-MmpE, and Comp-MmpE ^{Δ NLS1-2}, and systematically assessed bacterial persistence, pulmonary inflammation, and cytokine dynamics. Our results show that mice infected with the Δ MmpE strain exhibited significantly lower bacterial loads in the infected mice lung compared to those infected with the WT or Comp-MmpE strains. Remarkably, mice infected with the Comp-MmpE ^{Δ NLS1-2} strain showed significantly lower than those in the Comp-MmpE strains but higher than those in the Δ MmpE strains (**Figure 6A**). A similar trend was observed in splenic bacterial colonization (**Figure S7**). Hematoxylin and eosin (H&E) staining revealed extensive inflammatory infiltrates in the lungs of mice infected with WT or Comp-MmpE, whereas mice infected with Δ MmpE or Comp-MmpE ^{Δ NLS1-2} exhibited markedly reduced inflammation (**Figure 6B**). Mice infected with Δ MmpE and Comp-MmpE ^{Δ NLS1-2} strains exhibited significantly elevated levels of *Il1a*, *Il1b*, and *Il6* compared to those infected with WT or Comp-MmpE strains, despite lower bacterial burdens (**Figure 6C** and **E**).

Collectively, these results demonstrate that MmpE indeed contributes to *M. bovis* BCG survival *in vivo*, with its nuclear translocation being essential for mycobacterial intracellular survival.

Discussion

MmpE Functions as a Bifunctional Protein with Nuclear Localization and Phosphatase Activity

In this study, we identified two independent NLSs within the MmpE protein, RRR^{20–22} (NLS1) and RRK^{460–462} (NLS2), which together form a bipartite NLS motif that is rarely observed in bacterial effector proteins. Loss of a single NLS significantly reduced the nuclear accumulation of MmpE, with NLS2 deletion causing a more pronounced effect than that of NLS1. Simultaneous deletion of both motifs completely abolished nuclear localization (**Figure 1C–E**). These findings indicate that NLS1 and NLS2 function in a synergistic manner, with NLS2 likely playing a dominant role in nuclear import. Further sequence analysis revealed that NLS1 (RRR^{20–22}) is embedded within the conserved twin-arginine translocation (Tat) signal motif (S/TRR-X-FLK), a well-characterized determinant of Tat-dependent secretion (Gimenez *et al.*, 2018). The adjacent twin-arginine residues are critical for recognition by the Tat translocase, and mutations in this motif have been shown to significantly impair or abolish secretion (Ulfig *et al.*, 2017). Accordingly, the Comp-MmpE ^{Δ NLS1} strain likely exhibits impaired secretion, and the Comp-MmpE ^{Δ NLS2} strain is deficient in nuclear localization. Consistent with this, both the Comp-MmpE ^{Δ NLS1} and Comp-MmpE ^{Δ NLS2} strains exhibited lower intracellular survival than the Comp-MmpE strain and higher than that of the Δ MmpE strain (**Figure 3A–D**), suggesting that the nuclear localization ability of MmpE indeed contributes to bacterial survival. Interestingly, MmpE also functions as a Fe³⁺/Zn²⁺-dependent metallophosphatase, with residues H³⁴⁸ and N³⁵⁹ being essential for its catalytic

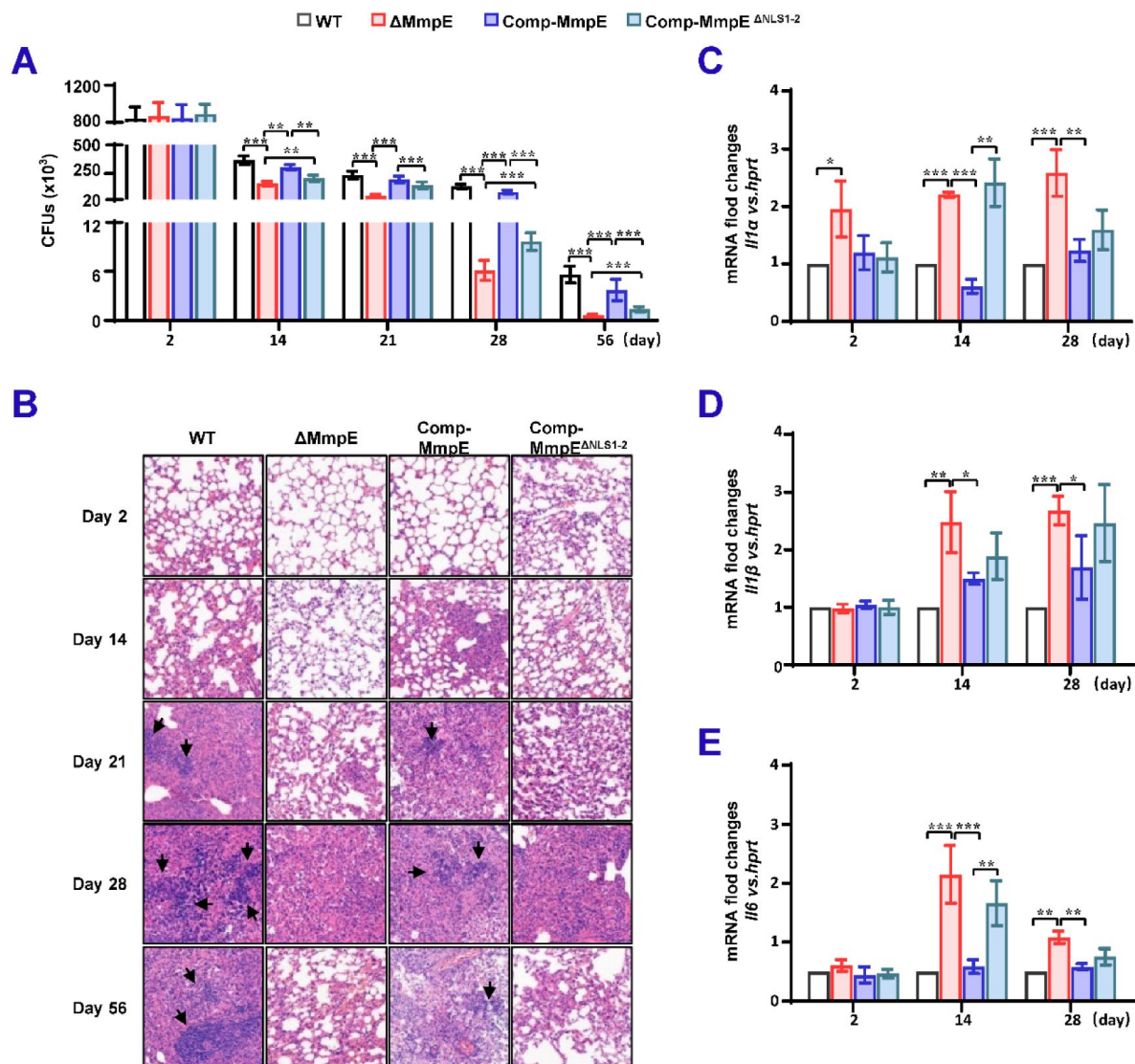


Figure 6.

Nuclear translocation of MmpE is essential for mycobacterial survival in mice.

(A) Bacterial burden in lungs of infected mice. Specific pathogen-free (SPF) C57BL/6 mice ($n = 6$ per group) were intratracheally infected with 1.0×10^7 colony-forming units (CFU) of *M. bovis* BCG strains, including wild-type (WT), MmpE deleted (Δ MmpE), MmpE-complemented (Comp-MmpE), or MmpE-complemented with NLS-deleted variants (Comp-MmpE Δ NLS1-2). Bacterial titers in lung homogenates were quantified by CFU assays at 0, 14, 28, and 56 days post-infection.

(B) Histopathology of infected lung tissues. Hematoxylin and eosin (H&E)-stained lung sections from mice infected as in (A) show granulomatous inflammation, Scale bars: 200 μ m.

(C-D) Pro-inflammatory cytokine expression in the spleen of infected mice. qRT-PCR analysis of cytokine mRNA levels of *Il1 α* (C), *Il1 β* (D) and *Il6* (E) in spleen tissues from infected mice ($n = 6$ /group) at 2 to 28 days post-infection.

Data are presented as mean $\pm$ SD from six biologically independent experiments. Statistical significance was determined by a two-tailed unpaired Student's *t*-test (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$).

activity (**Figure 2D** and **E**). The mutant strain (Comp-MmpE-H³⁴⁸AN³⁵⁹A) showed markedly reduced intracellular survival compared to Comp-MmpE (**Figure 3E**), indicating that the phosphatase activity of MmpE is also required for intracellular survival. This explains why the intracellular survival of the Comp-MmpE^{ANLS1-2} strain is significantly lower than that of the Comp-MmpE strain, yet obviously higher than that of the Δ MmpE strain (**Figure 3C** and **D**, **Figure 6A**).

Taken together, these findings suggest that MmpE is a dual-function effector protein that promotes intracellular survival during infection through the coordinated action of its nuclear localization signal and phosphatase activity.

MmpE's direct targeting of the host gene *VDR* represents a unique immune evasion strategy

VDR is a critical nuclear receptor that controls the expression of numerous genes involved in host immune defense, including antimicrobial peptides and cytokines (Usategui-Martín *et al.*, 2022). For example, Mtb suppresses the expression of the host gene *CYP27B1* to reduce active vitamin D synthesis, thereby limiting VDR activation and downstream antimicrobial responses (Liu *et al.*, 2006). Similarly, *Salmonella* downregulates VDR through miRNA-mediated mechanisms, weakening mucosal immunity (Chen *et al.*, 2014). In this study, we found that MmpE directly binds to the promoter of the *VDR* gene (**Figure 5F–J**) and attenuates the expression of VDR-regulated inflammatory genes (**Figure 3F**; **Figure 6C–E**), ultimately promoting bacterial intracellular survival (**Figure 6A**). This mode of action represents a novel strategy employed by bacterial pathogens to evade host defenses through the direct manipulation of host gene promoters by nucleomodulins. Although previous studies have reported that bacterial nucleomodulins interfere with the host transcriptional machinery, these effects have typically occurred indirectly, such as through epigenetic modifications or inhibition of signaling pathways (Jose *et al.*, 2016; Singh *et al.*, 2023). Direct promoter binding by a nucleomodulin, however, remains relatively uncommon and represents a distinct mechanism of host gene regulation.

MmpE regulates the expression of host genes involved in phagosome maturation

As an intracellular pathogen, Mtb employs multiple strategies to evade phagolysosomal degradation, a process essential for bacterial clearance (Chandra *et al.*, 2022; Zhang *et al.*, 2023; Singh and Nagaraja, 2025). Importantly, approximately 70% of phagosomes harboring Mtb fail to fuse with lysosomes, in part due to the activity of secreted phosphatases such as PtpA, PknG, and SapM, which disrupt vesicular trafficking (Wong *et al.*, 2011; Ge *et al.*, 2022; Zhang *et al.*, 2024; Alsayed and Gunosewoyo, 2024). MmpE has also been shown to contribute to the arrest of phagosome maturation in human cells, although the underlying mechanisms remain unclear (Forrellad *et al.*, 2020). In this study, we identify a potential mechanism by which MmpE inhibits phagosome maturation. Deletion of *mmpE* led to significant upregulation of lysosome-associated genes, including *TFEB*, *LAMP1*, *LAMP2*, and V-ATPase subunit genes (**Figure 4C** and **G**). Among these, TFEB is a master transcriptional regulator of lysosomal biogenesis and function (Jeong *et al.*, 2021; Wang *et al.*, 2024). Furthermore, ChIP-seq analysis revealed that MmpE directly binds to promoter regions of host genes within the PI3K–Akt–mTOR signaling pathway (**Figure S5A**), a key upstream regulator of TFEB activity (Fang *et al.*, 2021; Gounis *et al.*, 2025). Collectively, these results indicate that MmpE may act as a transcriptional regulator that targets the PI3K–Akt–mTOR–TFEB axis, thereby inhibiting lysosomal maturation.

MmpE is a $\text{Fe}^{3+}/\text{Zn}^{2+}$ -dependent metallophosphatase that is essential for mycobacterial intracellular survival

Previous studies have shown that phosphatases play critical roles in host–pathogen interactions by modulating phosphorylation-dependent signaling pathways (Sajid *et al.*, 2015 [↗](#); Bonne Køhler *et al.*, 2020 [↗](#)). In bacterial pathogens, these enzymes frequently contribute to immune evasion and intracellular survival. For example, the tyrosine phosphatase YopH from *Yersinia enterocolitica* disrupts phagocytic signaling in host macrophages, thereby impairing oxidative burst responses and promoting bacterial persistence (Davicino *et al.*, 2017 [↗](#); Shaban *et al.*, 2020 [↗](#)). Similarly, PpzA, a phosphatase from *Aspergillus fumigatus*, has been implicated in fungal adaptation to immune pressure under iron-replete conditions (Manfiolli *et al.*, 2017 [↗](#); Kurucz *et al.*, 2018 [↗](#)). In this study, we found that MmpE contains a predicted purple acid phosphatase (PAP)-like domain and exhibits phosphatase activity that requires Fe^{3+} and Zn^{2+} ions as cofactors (**Figure 2A** [↗](#) and **D** [↗](#)). Through site-directed mutagenesis, we identified His³⁴⁸ and Asn³⁵⁹ as critical residues for Fe^{3+} coordination. Substitution of these residues (H³⁴⁸AN³⁵⁹A) significantly impaired phosphatase function in the presence of Fe^{3+} , leading to a marked reduction in bacterial intracellular load (**Figure 2E** [↗](#); **Figure 3E** [↗](#)). However, the precise molecular mechanism by which MmpE interacts with Fe^{3+} remains to be elucidated in future studies.

In conclusion, our study identifies MmpE as a novel bifunctional virulence factor of mycobacteria, combining $\text{Fe}^{3+}/\text{Zn}^{2+}$ -dependent phosphatase activity with nuclear translocation capability. MmpE translocates into the host nucleus and directly binds to the promoters of some key host genes, such as *VDR*, thereby inhibiting the expression of the downstream targets implicated with the inflammation responses and lysosomal maturation, thereby promoting mycobacterial survival in macrophages and in mice (**Figure 7** [↗](#)). These findings reveal a sophisticated mechanism by which bacterial pathogen manipulates host gene transcription and signaling pathway to promote intracellular persistence.

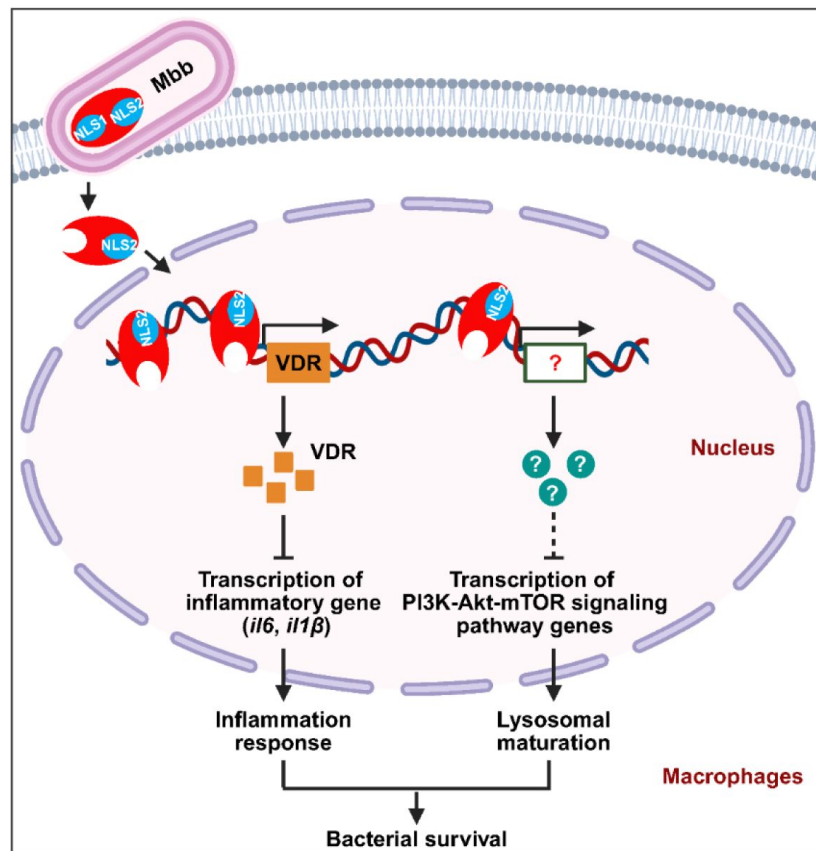


Figure 7.

Schematic diagram of nucleomodulin MmpE-mediated immune suppression and mycobacterial survival.

Upon mycobacterial infection, MmpE translocates into the host cell nucleus via its NLS2 motif (RRK⁴⁶⁰⁻⁴⁶²), where it binds to the promoter region of the *VDR* gene, repressing transcription of downstream inflammatory genes. Simultaneously, MmpE inhibits the PI3K-Akt-mTOR signaling pathway, leading to impaired phagosome maturation. These combined actions facilitate immune evasion and promote enhanced mycobacterial survival within macrophages and in mice.

Materials and methods

Key resources used in this study

Reagent or resource	Source	Identifier
Antibodies		
Anti- β -actin	Abbkine	Cat#ABL1010
Anti-Histone H3	Abbkine	Cat#ABL1070
Goat anti-rabbit IgG	Abbkine	Cat#A21020
Goat anti-mouse IgG	Abbkine	Cat#A25012
Anti-GFP	Proteintech	Cat#66002-1-Ig
Anti-Flag	Proteintech	Cat#66008-4-Ig
Cell and Mouse		
HEK293T	ATCC	Cat#CRL-3216
RAW264.7	ATCC	Cat#TIB-71
THP-1	ATCC	Cat#TIB-202
Mouse (C57BL/6)	Chang-sheng Bio (Liaoning, China)	
The plasmids used in this study are listed in Table S4		
The bacterial strains used in this study are listed in Table S5		
The primers used in this study are listed in Table S6		
Chemicals, Peptides, and Recombinant Proteins		
DMEM	Gibco	Cat#11965092
RPMI 1640	Gibco	Cat#C11875500BT
Opti-MEM™	Gibco	Cat#31985070
FBS	Gibco	Cat#A5256701
Sodium pyruvate	Gibco	Cat#11360070
L-glutamine	Gibco	Cat#A2916801

Chemicals, Peptides, and Recombinant Proteins

HEPES	Gibco	Cat#15630080
2-Mercaptoethanol	Gibco	Cat#21985023
penicillin-streptomycin antibiotics	Gibco	Cat#15140122
PMA	Sigma-Aldrich	Cat#P8139
<i>p</i> -NPP	Sigma-Aldrich	Cat#4264-83-9
Formaldehyde	Sigma-Aldrich	Cat#252549
OADC	MilliporeSigma	Cat#M0678
DAPI	Beyotime	Cat# P0126
Triton-X-100	Beyotime	Cat#P0096
RIPA buffer	Beyotime	Cat#P0038
NP-40	Beyotime	Cat#P0013F
protease inhibitor cocktail	Beyotime	Cat#AR1182
HieffTrans® Liposomal Transfection Reagent	YEASEN	Cat#40802ES03
protein A/G magnetic beads	MedChemExpress	Cat#HY-K0202

Critical commercial assays

TRIpure Reagent	Aidlab	Cat#RN0101
EASYspin RNA Mini Kit	Aidlab	Cat#RN0702
EndoFreePlasmidMiniKit	Aidlab	Cat#PL0401
cDNA Reverse Transcription Kit	Vazyme	Cat#R333-01
2 × ChamQ Universal SYBR qPCR Master Mix	Vazyme	Cat#Q711-03
Uniclone One Step Seamless Cloning Kit	Genesand	Cat#SC612
Clarity™ Western ECL Substrate	BIO-RAD	Cat#1705060
Nuclear and Cytoplasmic Protein Extraction kit	Beyotime	Cat#P0027

Software and algorithms

jvenn	https://bioinfogp.cnb.csic.es/tools/venny/index.html
AlphaFold v2.2.0	https://github.com/deepmind/alphafold
iTOL	https://itol.embl.de/
ESPrpt 3.0	https://esprpt.ibcp.fr/ESPrpt/cgi-bin/ESPrpt.cgi
GraphPad Prism 8	https://www.graphpad.com/
UCSF Chimera	http://www.cgl.ucsf.edu/chimera/
SignalP 5.0	https://services.healthtech.dtu.dk/services/SignalP-5.0/

Software and algorithms

MEGA 12.0	https://www.megasoftware.net/
STRING 12.0	https://cn.string-db.org/
Cytoscape 3.10.3	https://cytoscape.org/
CaseViewer version 2.0	https://www.3dhistech.com/solutions/caseviewer/

Bacterial strains and culture conditions

The bacterial strains used in this study are detailed in Table S5. *E. coli* DH5α and BL21 were cultured in Luria-Bertani (LB) medium under standard conditions. *M. bovis* BCG strains were grown in Middlebrook 7H9 broth supplemented with 10% (v/v) OADC (oleic acid, albumin, dextrose, catalase), 0.05% (v/v) Tween-80, and 2% (v/v) glycerol. Antibiotics were added as required at the following final concentrations: 50 µg/mL hygromycin (for mycobacteria) and 50 µg/mL kanamycin (for both mycobacteria and *E. coli*).

Cell culture

THP-1 cells were cultured in Roswell Park memorial institute 1640 (RPMI 1640) medium supplemented with 10% (v/v) heat-inactivated fetal bovine serum (FBS), 1 mM sodium pyruvate, 2 mM L-glutamine, 10 mM HEPES buffer (pH 7.2–7.5), and 50 µM 2-mercaptoethanol. For differentiation into macrophages, THP-1 cells were treated with 200 nM phorbol 12-myristate 13-acetate (PMA) for 24 h, followed by an additional 48 h incubation in fresh PMA-free medium to allow complete differentiation prior to experimental use.

HEK293T and RAW264.7 monocytes were cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% (v/v) FBS and 50 µg/mL penicillin-streptomycin. All cells were maintained at 37 °C in a humidified atmosphere with 5% CO₂.

Recombinant expression and secretion of MmpE

Recombinant *M. bovis* BCG strains expressing C-terminally 2×Flag-tagged Ag85B (positive control) and MmpE were constructed to assess protein expression and secretion under standard *in vitro* culture conditions. The strains were cultured in 200 mL of Middlebrook 7H9 broth supplemented with 10% (v/v) OADC, 2% (v/v) glycerol, and 0.05% Tween-80 at 37 °C until mid-log phase (OD₆₀₀ = 0.6). Cultures were centrifuged at 4,000 × g for 60 min to separate bacterial pellets from supernatants. Pellets were washed and lysed by bead beating in phosphate-buffered saline (PBS) supplemented with protease inhibitors. Culture supernatants were filtered through 0.22 µm sterile filters to remove residual bacteria and concentrated to a final volume of 2 mL using Amicon Ultra centrifugal filters (10 kDa molecular weight cutoff). Both bacterial lysates and concentrated supernatants were analyzed by SDS–PAGE followed by immunoblotting using anti-Flag antibodies to detect the expression and secretion of 2×Flag-tagged proteins.

Phylogenetic and sequence analysis

To investigate the evolutionary conservation and sequence features of MmpE homologs, phylogenetic reconstruction was performed using MEGA version 12.0 and iTOL platform with a neighbor-joining algorithm. Parameters included pairwise deletion for gap treatment, a poisson substitution model (selected for its suitability in analyzing closely related bacterial sequences), and 1,000 bootstrap replicates to assess node support. Only nodes with bootstrap values ≥50% were retained, and no additional tree pruning was applied.

The analysis combined metallophosphatase MmpE homologs from 11 *Mycobacterium* strains, including *M. tuberculosis* (Mtu), *M. bovis* BCG (Mbb), *M. orygis* (Mory), *M. kansasii* (Mku), *M. paratuberculosis* (Mpaa), *M. farcinogenes* (Mfg), *M. mucogenicum* (Mmuc),

M. vicinigordoniae (Mgor), *M. lentiflavum* (Mlw), *M. avium* (Mav), *M. manitobense* (Mman). Genomic sequences were retrieved from the NCBI RefSeq database, filtered for completeness, and aligned using Clustal Omega v1.2.4 with default parameters (gap opening penalty = 10, gap extension = 0.2, and iterative refinement enabled). Sequence conservation was visualized using ESPript 3.0, with residues exhibiting $\geq 90\%$ identity across all homologs highlighted in red.

Cell transfection and confocal microscopy

HEK293T cells were seeded at 70% confluency in poly-L-lysine-coated 35-mm glass-bottom dishes and maintained in DMEM supplemented with 10% (v/v) FBS at 37 °C under 5% CO₂. Transfection was performed using the Hieff Trans™ Liposomal Transfection Reagent according to the manufacturer's protocol. Briefly, 0.5 µg of plasmid DNA and 1.5 µL of transfection reagent were diluted in 50 µL of Opti-MEM, incubated at room temperature for 20 minutes, and then added dropwise to each well of 35-mm six-chamber glass-bottom dishes.

At 24–48 hpt, cells were washed with PBS, fixed in 4% paraformaldehyde for 15 min, and permeabilized with 0.1% Triton X-100 for 10 min. Nuclei were counterstained with 1 µg/mL DAPI (4',6-diamidino-2-phenylindole) for 10 min. Confocal imaging was performed using an Olympus FV1000 Confocal Laser Scanning Microscope equipped with a 100×/1.40 NA oil immersion objective. EGFP and DAPI signals were detected with 488 nm and 405 nm lasers, respectively. Spectral unmixing was carried out using the Olympus FV10-ASW software (v4.2) to minimize signal cross-talk. Z-stack images were acquired in sequential mode with a 0.5 µm step size across all samples to ensure imaging consistency.

Immunoblot analysis

For immunoblotting, the cells were lysed in radioimmunoprecipitation assay (RIPA) buffer supplemented with 1.5% (v/v) protease inhibitor cocktail. Protein concentrations were quantified using the bicinchoninic acid (BCA) assay. Equal amounts of protein (20–40 µg) were resolved by SDS-PAGE and transferred onto PVDF membranes. Membranes were blocked for 1 h at room temperature with 5% (w/v) non-fat milk in TBST (25 mM Tris, 150 mM NaCl, 2 mM KCl, and 0.2% Tween-20, pH 7.4) and incubated overnight at 4 °C with primary antibodies. After three washes with TBST (10 min each), membranes were incubated with horseradish peroxidase (HRP)-conjugated secondary antibodies for 1 hour at room temperature. Protein bands were visualized using an enhanced chemiluminescence (ECL) substrate and imaged using the ChemiDoc™ MP Imaging System (Bio-Rad). β -actin, GAPDH, or histone H3 was used as an internal control for normalization. Antibody details are provided in the Key resources table.

For subcellular fractionation, cytoplasmic and nuclear proteins were isolated using a Nuclear and Cytoplasmic Protein Extraction Kit (manufacturer's protocol). Briefly, after treatment, cells were washed and harvested in cold PBS. Cytoplasmic and nuclear fractions were separated according to the manufacturer's instructions. Protein concentrations were measured using the BCA assay, and equal amounts of protein were analyzed by immunoblotting as described above.

AlphaFold Predicted Structural Modeling of MmpE with NLS Truncations

The structural models of MmpE with NLS truncations were generated using AlphaFold version 2.2.0. The full-length MmpE sequence was used as the input to predict the reference structure. To assess the impact of NLS deletions, modified sequences were prepared by removing the residues corresponding to NLS1 (RRR^{20–22}), NLS2 (RRK^{460–462}), or both motifs. Structural prediction was performed for each sequence using the default AlphaFold parameters, which include the application of the pre-trained model and multiple sequence alignment (MSA) generation using the UniRef90 and MGnify databases. The confidence of the predictions was evaluated using the

predicted local distance difference test (pLDDT) scores provided by AlphaFold. The resulting models were visualized and analyzed using UCSF ChimeraX to compare the overall protein conformations and assess structural changes due to NLS deletions.

Prediction of metal ion-binding residues in MmpE

The structure of MmpE was predicted using AlphaFold version 2.2.0 to investigate its potential metal ion dependence and identify key residues involved in metal coordination. The predicted structure was analyzed using PyMOL to identify residues within the putative active site. Residues were selected based on their spatial arrangement in the active site, and their proximity to regions likely involved in metal ion binding was annotated. Visualization and annotation of the predicted active site were performed using PyMOL to determine the potential roles of these residues in metal ion coordination.

Protein Expression and Purification

N-terminal maltose-binding protein (MBP)-tagged and C-terminal His₆-tagged wild-type MmpE, as well as mutant variants lacking the Tat signal peptide were expressed in *E. coli* BL21 (DE3). Protein expression was performed in 1 L of LB medium. Cultures were grown at 37 °C until the optical density at 600 nm (OD₆₀₀) reached approximately 0.8, after which protein expression was induced with 0.2 mM isopropyl-β-D-thiogalactopyranoside (IPTG) at 18 °C for 24 h. Following induction, cells were harvested by centrifugation and lysed by sonication in ice-cold lysis buffer containing 150 mM NaCl, 30 mM Tris-HCl (pH 7.5), and 1 mM phenylmethylsulfonyl fluoride (PMSF). The lysate was clarified by centrifugation, and the supernatant was subjected to affinity purification using an MBP affinity column.

Phosphatase activity assay

Phosphatase activity was measured using *p*-nitrophenyl phosphate (*p*-NPP) as the substrate in a final reaction volume of 200 μL. Reactions were initiated by incubating 0–80 μg of purified protein with *p*-NPP at 37 °C for 60 min. The release of *p*-nitrophenol (*p*-NP), the dephosphorylated product, was quantified by measuring absorbance at 405 nm using a microplate spectrophotometer.

Kinetic parameters were determined under initial velocity (V_0) conditions using substrate concentrations ranging from 0.1 to 10 mM *p*-NPP. The kinetic constants K_m and V_{max} were calculated by fitting the data to the Michaelis-Menten equation using nonlinear regression analysis in GraphPad Prism 8.0. The catalytic turnover rate (K_{cat}) was calculated using the equation $K_{cat} = V_{max}/E_0$, where V_{max} was expressed as ηM *p*-NP min⁻¹ and E_0 represents the active enzyme concentration in ηM . Control reactions were carried out in the absence of either the substrate or the enzyme to measure background absorbance. All assays were performed in triplicate to ensure the reproducibility of results.

Quantitative real-time PCR

THP-1 cells were infected with the *M. bovis* BCG strain and harvested 20 h post-infection for total RNA isolation. RNA was extracted using 1 mL of TRIzol[®] reagent following the manufacturer's protocol. To eliminate genomic DNA contamination, RNA samples were treated with DNase I for 30 min at 37 °C, followed by heat inactivation at 65 °C for 10 min. RNA concentration and purity were assessed using a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific).

For cDNA synthesis, 1 μg of total RNA was reverse transcribed using the HiScript[®] III cDNA Reverse Transcription Kit with random hexamers primers, according to the manufacturer's protocol. qRT-PCR was performed using ChamQ SYBR[®] Green RT-PCR Master Mix. Relative mRNA expression levels were calculated using the 2^{-ΔΔCt} method, normalizing Ct values to the

housekeeping gene *HPRT*. All qRT-PCR experiments were conducted in triplicate, with at least three independent biological replicates and two technical replicates per condition. Primer sequences are listed in Table S6.

RNA-seq analysis

THP-1 cells were seeded at a density of 1×10^6 cells well in 6-well plates and differentiated into macrophages by treatment with 100 nM PMA for 48 h. After differentiation, cells were washed with PBS and infected with *M. bovis* BCG WT or *KommpE* strains at a multiplicity of infection (MOI) of 10:1. Following 20 h of infection, total RNA was extracted using TRIzol[®] reagent according to the manufacturer's protocol.

High-quality sequencing reads were then aligned to the human reference genome (GRCh38) using HISAT2 (v2.2.1). Transcript quantification was performed using FeatureCounts (v2.0.3), and differential gene expression analysis was conducted using DESeq2 (v1.38.3) in R. Genes with an adjusted *P*-value < 0.05 and an absolute $|\log_2 \text{fold change}| \geq 1$ were considered significantly differentially expressed. RNA-seq data were visualized using GraphPad Prism version 8.0, as well as volcano plots, heatmaps, and other graphical outputs generated via the SRplot platform (Tang *et al.*, 2023 [DOI](#)).

CUT&Tag analysis

HEK293T cells were cultured in 10 cm² flat dish at a density of 1×10^7 cells per well and transfected with 10 µg of the MmpE-pEGFP expression plasmid using Hieff Trans[™] Liposomal Transfection Reagent, following the manufacturer's protocol. Cells transfected with the empty vector pEGFP-N1 served as controls. After 36 h of transfection, cells were harvested for CUT&Tag analysis.

For the CUT&Tag, cells were cross-linked with 1% formaldehyde at room temperature for 10 minutes, and the reaction was quenched with 0.125 M glycine. Cells were collected by centrifugation and permeabilized using ice-cold wash buffer containing 0.05% digitonin. Primary antibody incubation was performed overnight at 4 °C using mouse anti-GFP antibody (1:500 dilution), followed by a 2-hour incubation at room temperature with a Protein A/G-conjugated secondary antibody (1:100 dilution). The cells were then incubated at 37 °C for 1 hour with a Protein A–Tn5 transposase complex (1:100 dilution) preloaded with Illumina sequencing adapters, enabling site-specific cleavage and adapter insertion near MmpE binding sites.

The reaction was terminated using lysis buffer containing 10% SDS, followed by proteinase K digestion. DNA was purified and amplified by PCR (12 cycles) using the NEBNext Ultra II DNA Library Preparation Kit. Fragment size distribution (150–300 bp) was confirmed using the Agilent 2100 Bioanalyzer.

Sequencing was performed on the Illumina NovaSeq 6000 platform, producing 150 bp paired-end reads. Raw reads were quality-checked using FastQC and aligned to the human reference genome (hg38). Peak calling was conducted using MACS2 (v2.2.7) with the parameters `--nomodel --qvalue 0.05`, identifying 2,903 high-confidence MmpE binding sites with >9-fold enrichment. Genomic annotation, including promoter analysis and gene ontology (GO) enrichment, was performed using HOMER (v4.11). *De novo* motif discovery was also conducted with HOMER, and motifs were statistically validated using TOMTOM with a significance threshold of $P < 0.001$.

Chromatin Immunoprecipitation (ChIP) assays

ChIP assays were performed to validate the direct binding of MmpE to the promoter regions of target genes. HEK293T cells were transfected with either an empty control vector, EGFP, or MmpE–EGFP expression plasmids. Cross-linking, chromatin isolation, fragmentation, and

immunoprecipitation were carried out following the same procedure described for the CUT&Tag experiments.

DNA was purified from both input and immunoprecipitated (IP) samples and subjected to PCR using primers specific to the promoter regions of the target genes (–100 bp to +50 bp relative to the transcription start site). *GAPDH* promoter primers were used as a negative control. PCR products were analyzed by agarose gel electrophoresis to assess the presence of enriched promoter fragments.

For quantitative analysis, ChIP–qPCR was performed to measure the enrichment of specific promoter regions in IP samples. All reactions were carried out in triplicate. Relative enrichment was calculated using the $2^{-\Delta Ct}$ method, normalized to input DNA.

Electrophoretic mobility shift assays (EMSA)

DNA substrates were amplified by PCR from the genomic DNA of *Mycobacterium tuberculosis* H37Rv. The sequences of the oligonucleotide fragments used in the assay are listed in Table S6. EMSA was carried out using a modified protocol as previously described (Li *et al.*, 2020 [DOI](#)). Each 20 μ L reaction contained PCR-amplified DNA fragments, purified MmpE protein at varying concentrations, and binding buffer composed of 50 mM Tris–HCl (pH 7.5), 50 mM NaCl, 50 mM FeCl_3 , 0.05 mg/mL bovine serum albumin (BSA), and 10% (v/v) glycerol. Reaction mixtures were incubated on ice for 30 minutes to allow protein–DNA complex formation. Samples were then resolved on a 4.8% native polyacrylamide gel using a running buffer containing 15 mM Tris–HCl (pH 7.5) and 0.1 M glycine at 80 V. Following electrophoresis, gels were stained with ethidium bromide and visualized using a Gel Imaging Analysis System (JIAPENG, China).

Colony-Forming Unit (CFU) assay

For bacterial preparation, mid-log phase cultures of *M. bovis* BCG ($\text{OD}_{600} = 0.8$) were harvested, washed with PBS, and resuspended in fresh medium. THP-1 cells were seeded in 24-well plates at a density of 5×10^5 cells per well and infected at a MOI of 1:10. After 4 hours of infection, extracellular bacteria were removed by treatment with penicillin–streptomycin (100 μ g/mL) for 2 hours. Cells were then washed three times with PBS and maintained in antibiotic-free medium until harvest. To count intracellular bacteria, macrophages were lysed with 0.25% SDS for 10 minutes at room temperature. Lysates were serially diluted in PBS, and 100 μ L aliquots of each dilution were plated on Middlebrook 7H10 agar supplemented with 10% OADC and kanamycin (50 μ g/mL). Plates were incubated at 37 °C for 14 days. CFUs were enumerated, and final counts were calculated by applying the corresponding dilution factors. *M. bovis* BCG strains used in this study are listed in Table S5.

Animal experiments

Specific pathogen-free (SPF) female C57BL/6 mice (6–8 weeks old, 16–18 g) were obtained from Chang-sheng Bio (Liaoning, China). All animals were maintained under SPF conditions in individually ventilated cages with controlled temperature and humidity, on a 12-hour light/dark cycle. Mice had ad libitum access to sterilized food and water.

Mice infection

Mice were infected as described previously (Chen *et al.*, 2022 [DOI](#)), with minor modifications. Mid-log phase *M. bovis* BCG cultures were washed twice in PBS containing 0.05% Tween 80 and briefly sonicated to disrupt bacterial clumps. Female C57BL/6 mice were anesthetized with isoflurane and intranasally inoculated with 1×10^7 CFUs of *M. bovis* BCG in 40 μ L of PBS. To confirm the inoculation dose, six mice were euthanized at 48 hours post-infection. Lungs were harvested,

homogenized in 1 mL of PBS using a tissue homogenizer, and serial dilutions of the homogenates were plated on Middlebrook 7H10 agar supplemented with 10% OADC. Plates were incubated at 37 °C with 5% CO₂ for 14 days, after which CFUs were enumerated.

Histological analysis

Lung tissues from *M. bovis* BCG-infected mice were fixed in 4% phosphate-buffered formalin at room temperature for 24 h, followed by paraffin embedding. Paraffin-embedded tissue samples were sectioned into 2–3 µm thick slices using a microtome. Sections were deparaffinized, rehydrated through a graded ethanol series, and stained with hematoxylin and eosin (H&E). Stained tissue sections were examined using an Olympus BX53 light microscope. For digital analysis and annotation, scanned slides were visualized using CaseViewer software (version 2.0; 3DHISTECH).

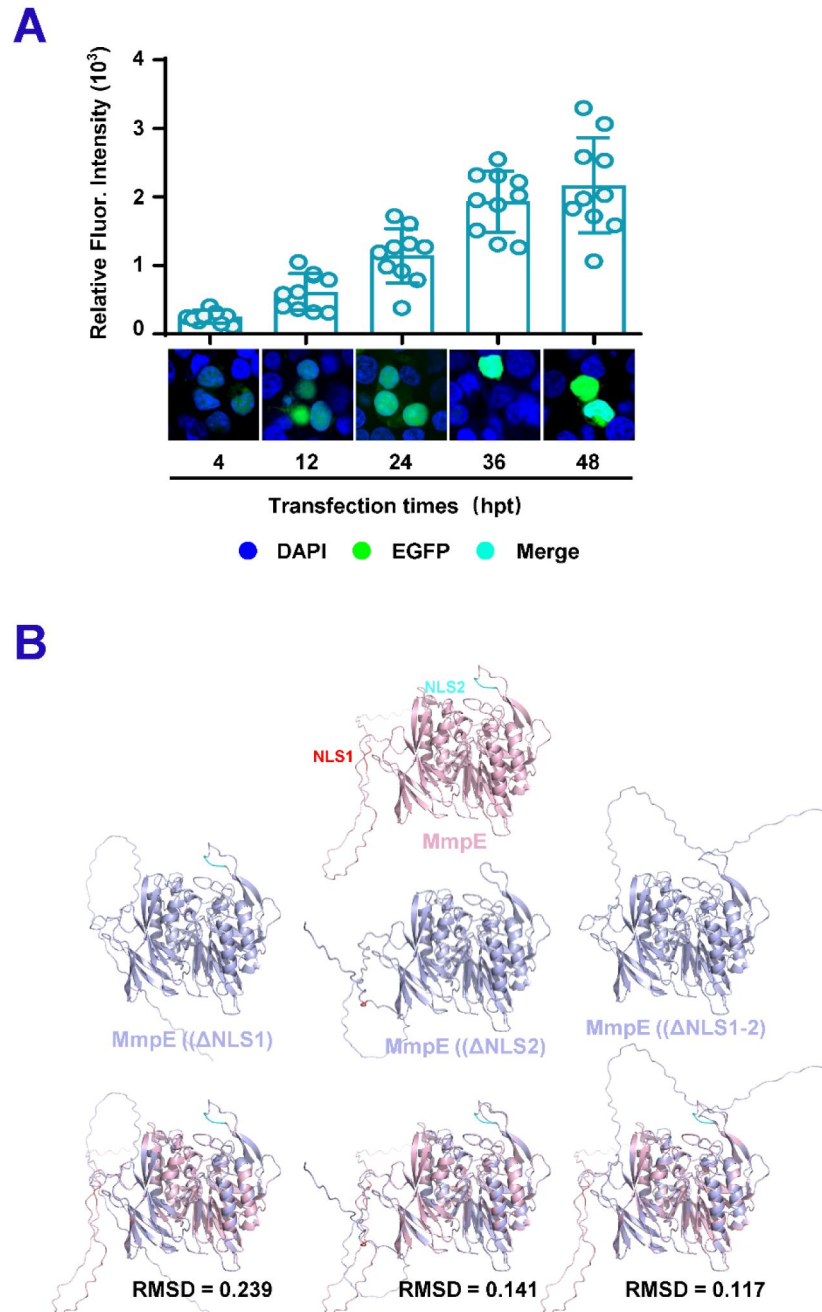
Ethics approval

All animal experiments were conducted in accordance with the protocols approved by the Animal Ethics Committee of the Huazhong Agricultural University.

Statistical analyses

Data are presented as mean ± standard deviation (SD). Statistical comparisons between two groups were performed using two-tailed unpaired Student's *t*-tests, following verification of normal distribution. A *P*-value less than 0.05 was considered statistically significant. Significance levels are indicated in figures as follows: *P* > 0.05 (ns, not significant), *P* < 0.05 (*), *P* < 0.01 (**), and *P* < 0.001 (***). All statistical analyses were performed using GraphPad Prism (version 8.0). Data were obtained from at least three independent biological replicates, each including two or more technical replicates.

Supplementary Figures



Supplementary Figure 1.

Identification of MmpE as a nucleomodulin in *Mycobacterium*.

(A) MmpE is a mycobacterial nucleomodulin. Subnuclear localization of MmpE-EGFP. Confocal microscopy was used to assess the nuclear distribution of MmpE-EGFP at various time points post-transfection. Nuclei were stained with DAPI (blue); MmpE-EGFP is shown in green. Scale bar: 10 μ m.

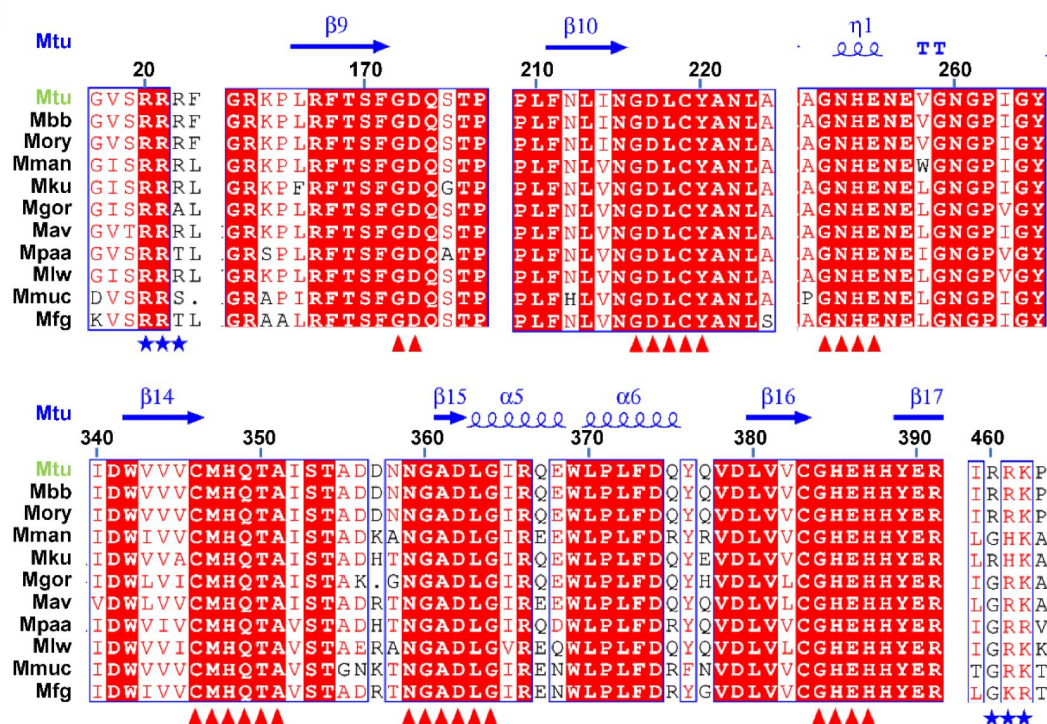
(B) AlphaFold-predicted structural models of MmpE with NLS truncations. Structural models of MmpE were generated using AlphaFold 2.2.0 to assess the impact of NLS deletions on overall protein conformation. (upper) Full-length MmpE structure shown as a reference. (middle) MmpE structure with NLS deleted. (botten) Predicted structures of MmpE with individual deletions of NLS1, NLS2, or both motifs. Confidence scores (pLDDT) were used to evaluate prediction reliability, and UCSF ChimeraX was used to visualize and compare structural changes.

A

Protein Chain	Description	Z Value	RMSD Value
6py9-C	Fe ⁽³⁺⁾ -Zn ⁽²⁺⁾ purple acid phosphatase	36.5	2.3
6py9-D	Fe ⁽³⁺⁾ -Zn ⁽²⁺⁾ purple acid phosphatase	36.5	2.3
6py9-A	Fe ⁽³⁺⁾ -Zn ⁽²⁺⁾ purple acid phosphatase	36.4	2.3
4kkz-C	purple acid phosphatase	36.3	2.3
4kkz-A	purple acid phosphatase	36.3	2.3
6g46-D	Fe ⁽³⁺⁾ -Zn ⁽²⁺⁾ purple acid phosphatase	36.3	2.3
2qfp-B	purple acid phosphatase	36.3	2.3

Z value>10 and RMSD value<4 indicate that the comparison result is extremely reliable

B

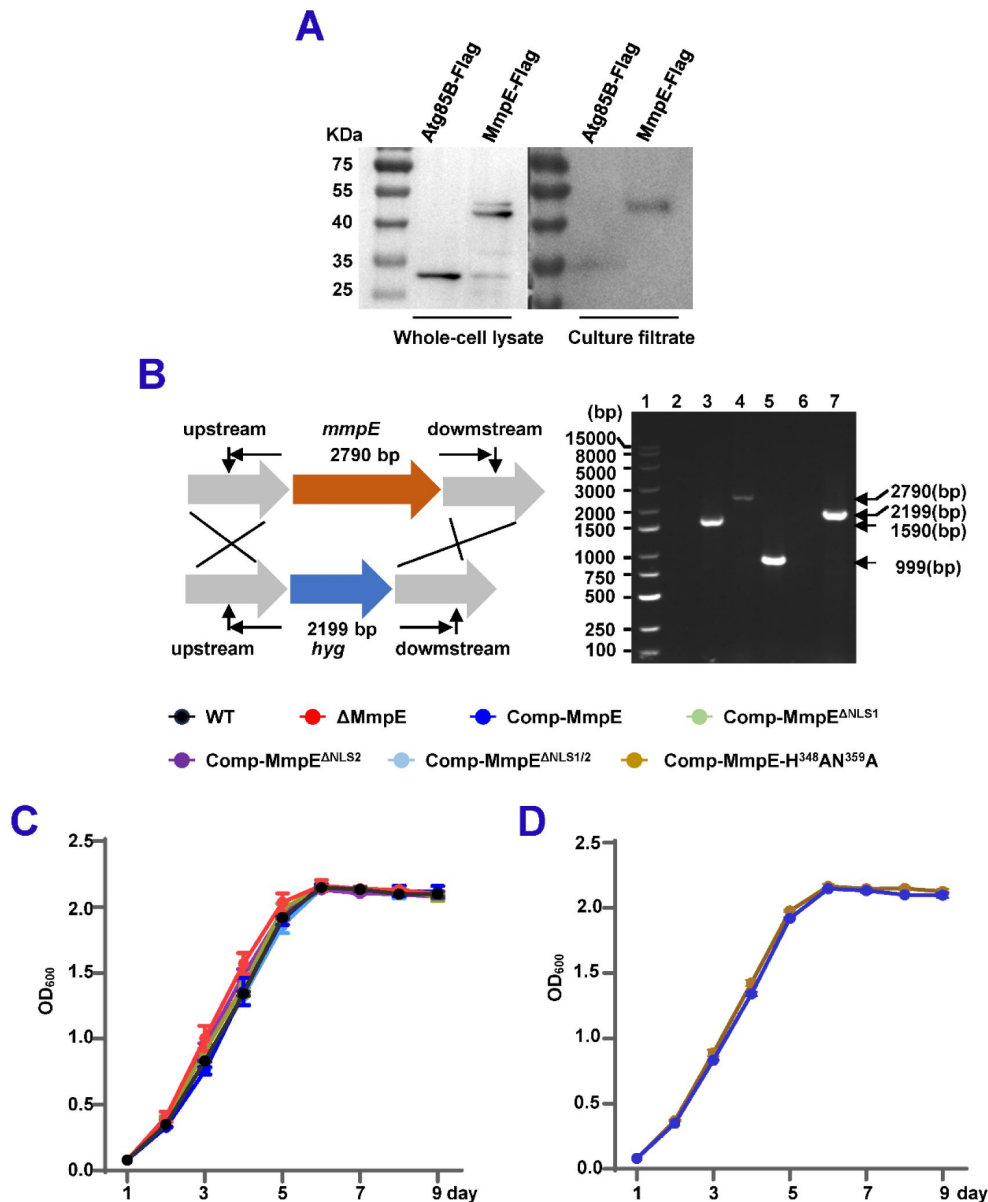


Supplementary Figure 2.

Identification of MmpE as a conserved Fe³⁺/Zn²⁺-metallophosphatase in Mycobacteria.

(A) Sequence and structure-based analyses of MmpE. Sequence and structure-based analyses were performed using UCSF Chimera to investigate the potential function and structural features of MmpE. The structural comparison results, with a Z value >10 and RMSD value <4, indicate high confidence in the predicted structural models.

(B) Clustal Omega sequence alignment of MmpE from various mycobacterial species, highlighting conserved residues ($\geq 90\%$ identity, shown in red). Blue stars indicate the predicted NLSs, and red triangles mark characteristic sequences found in metallophosphatases. Species abbreviations: *M. tuberculosis* (Mtu), *M. bovis* BCG (Mbb), *M. orygis* (Mory), *M. kubicae* (Mku), *M. paratuberculosis* (Mpaa), *M. farcinogenes* (Mfg), *M. mucogenicum* (Mmuc), *M. vicinagordoniae* (Mgor), *M. lentiflavum* (Mlw), *M. avium* (Mav), *M. manitobense* (Mman).



Supplementary Figure 3.

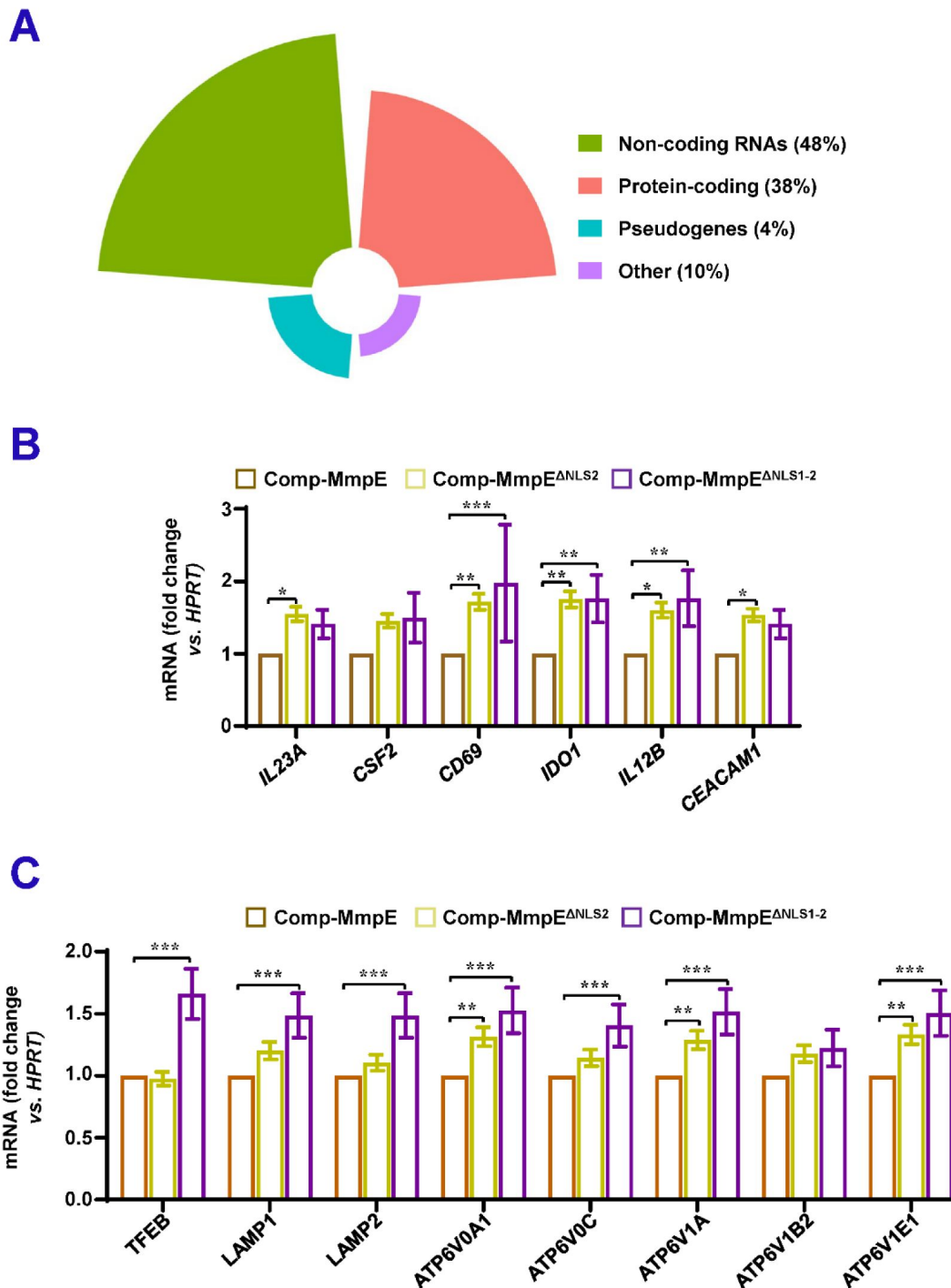
The nuclear translocation and phosphatase activity of MmpE are essential for *M. bovis* BCG survival in macrophage cells.

(A) Detection of mycobacterial MmpE secretion ability. C-terminally 2×Flag-tagged Ag85B (positive control) and MmpE were expressed in *M. bovis* BCG strains. Immunoblot analysis of bacterial lysates and culture supernatants showing the expression and secretion of Flag-tagged Ag85B and MmpE.

(B) Construction and validation of the MmpE-deleted strain of *M. bovis* BCG. (left) Schematic diagram of the homologous recombination strategy used to delete *mmpE* from the *M. bovis* BCG genome. (right) Wild-type and mutant strains were used as templates to amplify the *mmpE* gene (600 bp upstream–600 bp downstream) by PCR. Lanes 1 and 3: wild-type genomic DNA; lanes 2 and 4: $\Delta mmpE$ genomic DNA.

(C-D) Growth curve analysis of BCG strains. Growth of *M. bovis* BCG strains, including wild-type BCG (WT), MmpE-deleted ($\Delta mmpE$), MmpE-complemented (Comp-MmpE), and NLS-deleted complemented (Comp-MmpE $\Delta NLS1-2$) strains **(C)**, or phosphatase activity site mutation complement (Comp-MmpE-H³⁴⁸AN³⁵⁹H) **(D)**, was measured in 7H9 medium.

Data are presented as mean $\pm$ SD from three biologically independent experiments. Statistical significance was determined by a two-tailed unpaired Student's *t*-test.



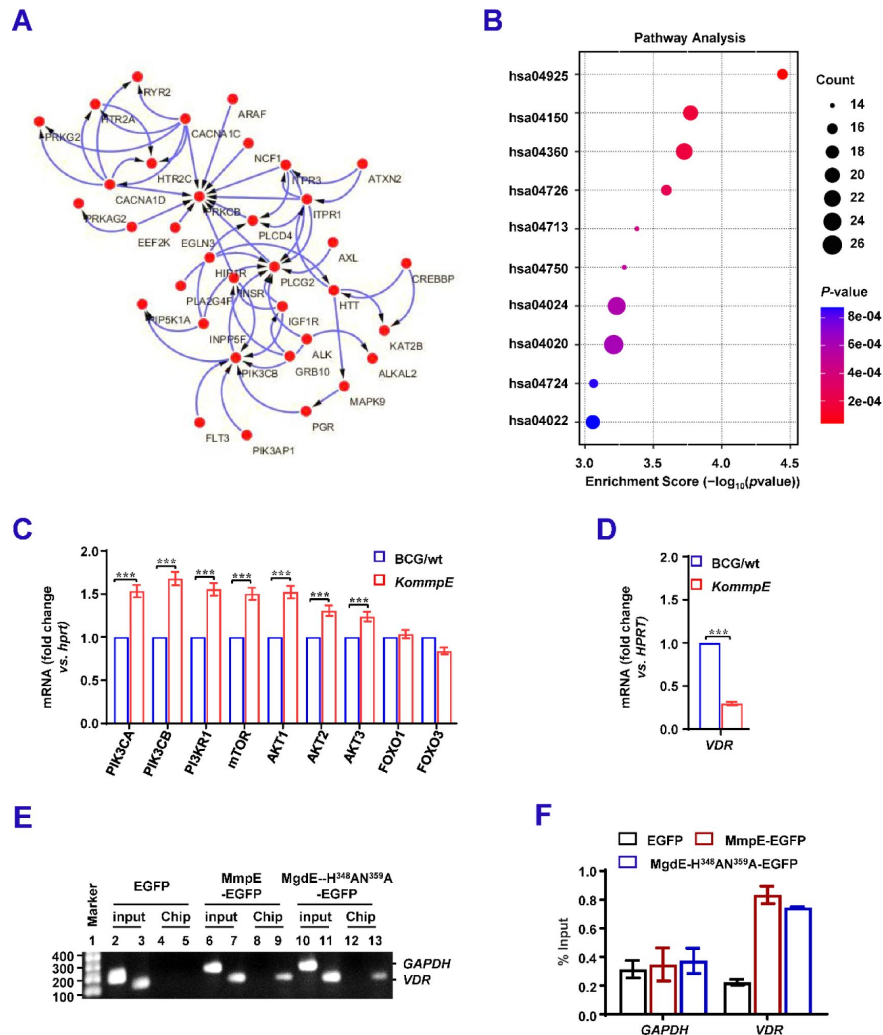
Supplementary Figure 4.

MmpE modulates host transcription network involved in inflammation response and lysosomal maturation

(A) Biotype distribution of potential MmpE-regulating DEGs genes in THP-1 cells.

(B-C) Quantitative RT-PCR analysis of gene expression in infected THP-1 cells. THP-1 cells were infected with complemented strain (Comp-MmpE) and NLS-deleted complemented (Comp-MmpE^{ANLS2}, Comp-MmpE^{ANLS1-2}) strains for 20 h. mRNA levels of cytokine genes (B); mRNA levels of genes involved in lysosomal acidification and biogenesis (C).

Data are presented as mean $\pm$ SD from three biologically independent experiments. Statistical significance was determined by a two-tailed unpaired Student's *t*-test (**P* < 0.05, ***P* < 0.01, ****P* < 0.001).



Supplementary Figure 5.

MmpE suppresses the expression of *VDR* and inhibits activation of the PI3K-Akt-mTOR signaling pathway.

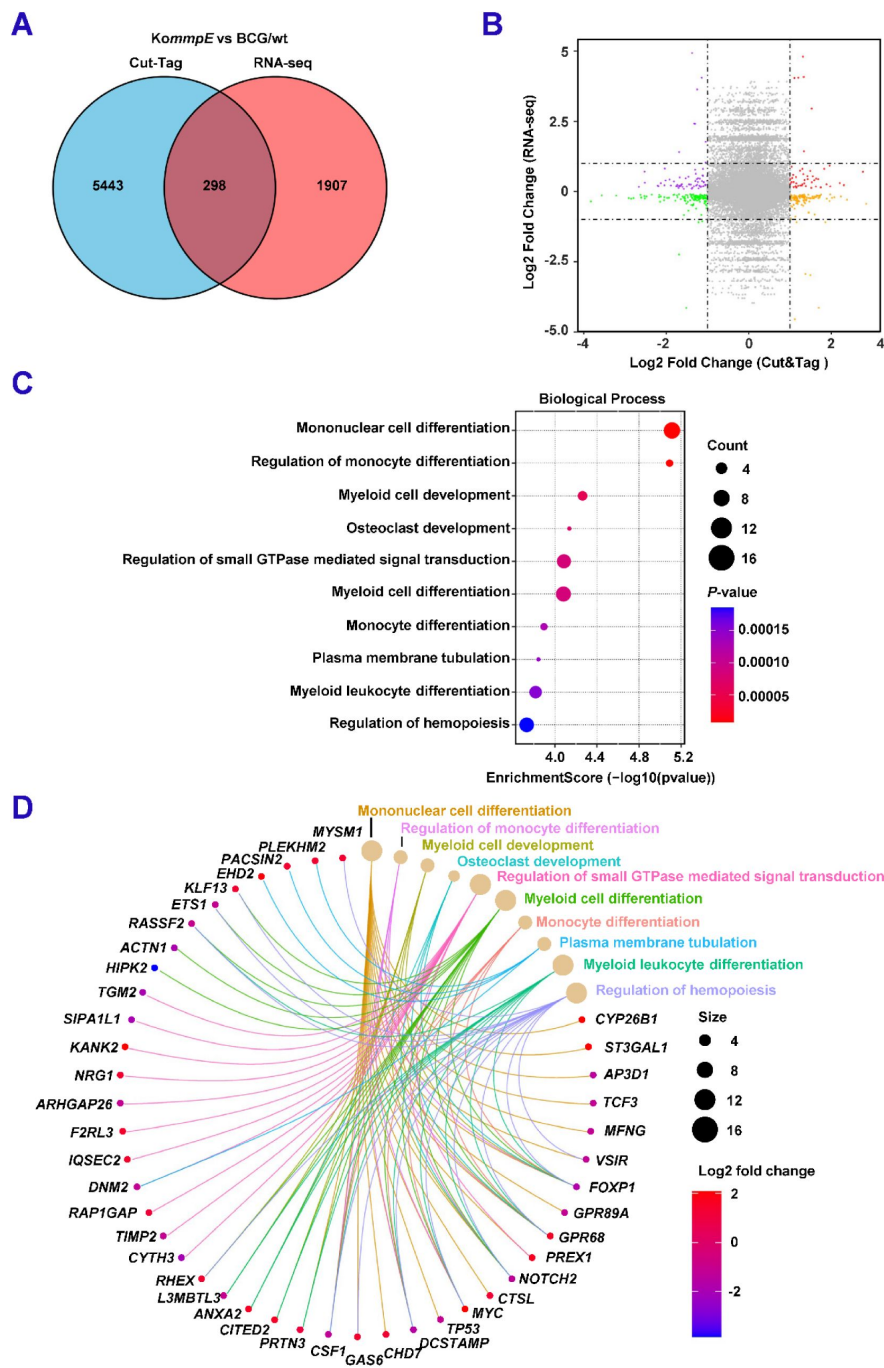
(A) Interaction network of DEGs associated with MmpE-binding sites. The network was constructed using STRING v12.0 and visualized with Cytoscape, highlighting interactions between genes potentially regulated by MmpE.

(B) KEGG pathway enrichment analysis of MmpE-binding sites identified in EGFP-MmpE-transfected HEK293T cells revealed significant enrichment in multiple signaling pathways, including aldosterone-regulated sodium reabsorption (hsa04925), mTOR signaling (hsa04150), axon guidance (hsa04360), serotonergic synapse (hsa04726), circadian entrainment (hsa04713), inflammatory mediator regulation of TRP channels (hsa04750), cAMP signaling (hsa04024), calcium signaling (hsa04020), glutamatergic synapse (hsa04724), and cGMP-PKG signaling (hsa04022). The bar plot shows selected significantly enriched pathways. Bar length represents the number of MmpE-binding sites associated with each pathway, and the color gradient indicates statistical significance, represented as $-\log_{10}(P\text{-value})$.

(C-D) Quantitative RT-PCR analysis of gene expression in infected THP-1 cells. THP-1 cells were infected with wild-type BCG (BCG/wt), MmpE-deleted (*KommpE*) strains for 20 h. mRNA levels of PI3K-Akt pathway (C); mRNA levels of *VDR* gene (D).

(E-F) ChIP-PCR and qPCR analyses of *VDR* promoter binding by MmpE and the phosphatase-deficient mutant MmpE-H³⁴⁸AN³⁵⁹A in HEK293T cells. Chromatin was immunoprecipitated from cells transfected with EGFP control, MmpE-EGFP, or MmpE-H³⁴⁸AN³⁵⁹A-EGFP constructs. PCR was performed using primers targeting the *VDR* promoter and *GAPDH* (negative control), and products were analyzed by agarose gel electrophoresis (E). ChIP enrichment for the *VDR* and *GAPDH* promoter regions was quantified by qPCR (F), with enrichment calculated using the $2^{-\Delta Ct}$ method.

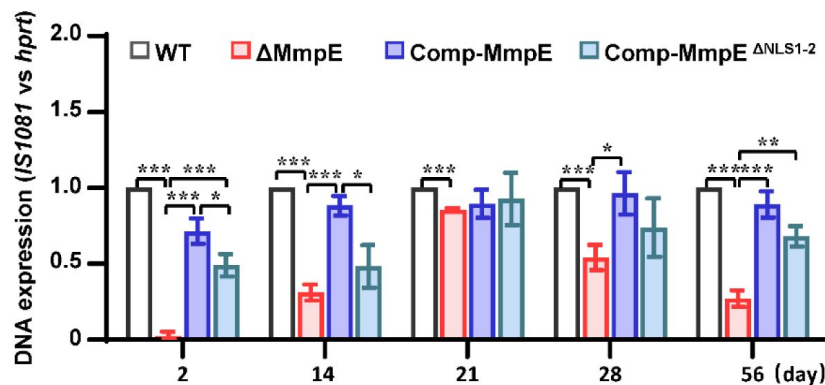
Data are presented as mean $\pm$ SD from three biologically independent experiments. Statistical significance was determined by a two-tailed unpaired Student's *t*-test (***P* < 0.001).



Supplementary Figure 6.

MmpE modulates the transcription of immune-associated genes.

- (A) Association between differentially expressed genes and differential peak-related genes. Venn diagram showing the overlap between differential peak-related genes identified by CUT&Tag sequencing and DEGs identified by RNA-seq analysis.
- (B) Four-quadrant scatter plot comparing CUT&Tag-seq and RNA-seq results. The $\log_2$ (fold change) of peak-related genes from CUT&Tag-seq is plotted against the $\log_2$ (fold change) of FPKM values from RNA-seq analysis. The scatter plot demonstrates the correlation between chromatin binding changes and gene expression levels.
- (C) GO enrichment analysis of DEGs revealed significant overrepresentation of Biological Process (BP) terms in macrophages infected with *KommpE* compared to those infected with wild-type BCG (BCG/wt).
- (D) Chord plot showing significantly altered genes associated with GO Biological Process pathways in *KommpE*-infected macrophages compared to BCG/wt infection.



Supplementary Figure 7.

MmpE promotes mycobacterial colonization in the spleen of mice.

Bacterial colonization in the spleen. CFU assays were performed on splenic homogenates from mice infected as described in **Figure 6A**. Bacterial loads (CFU/g tissue) were quantified at 2, 14, 21, 28, and 56 days post-infection. Data are presented as mean $\pm$ SD (n = 3). Statistical significance was determined using a two-tailed unpaired Student's *t*-test ($P < 0.05$, $**P < 0.01$, and $***P < 0.001$).

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Additional information

Author contributions

Conceptualization: L. C., A. G., and L. Z.; Methodology: L. C.; Validation: L. C., B. D., and Q. J.; Formal analysis: L. C., Y. W.; Investigation: L. C., B. D., Q. J., and Y. W.; Resources: A. G., L. Z., and Y. C.; Data curation: L. C., Q. J., and Y. W.; Writing - original draft: L. C., L. Z., and A. G.; Writing - review & editing: L. C., L. Z., and A. G.; Visualization: L. C., Q. J., Y. C., L. Z., and A. G.; Supervision: A. G., and L. Z.; Project administration: A. G., and L. Z.; Funding acquisition: A. G., L. Z., and Y. C.

Additional files

Table S1. 175 genes were differentially expressed genes in RNA-seq. [↗](#)

Table S2. 2903 candidate MmpE-specific ChIP-seq signals. [↗](#)

Table S3. 298 genes were differentially expressed in both CUT&Tag and RNA-seq. [↗](#)

Table S4. Plasmids used in this study. [↗](#)

Table S5. Bacterial strains used in this study. [↗](#)

Table S6. Primers used in this study. [↗](#)

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Reviewer #1 (Public review):

Summary:

Review of the manuscript titled "Mycobacterial Metallophosphatase MmpE acts as a nucleomodulin to regulate host gene expression and promotes intracellular survival".

The study provides an insightful characterization of the mycobacterial secreted effector protein MmpE, which translocates to the host nucleus and exhibits phosphatase activity. The study characterizes the nuclear localization signal sequences and residues critical for the phosphatase activity, both of which are required for intracellular survival.

Strengths:

- (1) The study addresses the role of nucleomodulins, an understudied aspect in mycobacterial infections.
- (2) The authors employ a combination of biochemical and computational analyses along with in vitro and in vivo validations to characterize the role of MmpE.

Weaknesses:

- (1) While the study establishes that the phosphatase activity of MmpE operates independently of its NLS, there is a clear gap in understanding how this phosphatase activity supports mycobacterial infection. The investigation lacks experimental data on specific substrates of MmpE or pathways influenced by this virulence factor.
- (2) The study does not explore whether the phosphatase activity of MmpE is dependent on the NLS within macrophages, which would provide critical insights into its biological relevance in host cells. Conducting experiments with double knockout/mutant strains and comparing their intracellular survival with single mutants could elucidate these dependencies and further validate the significance of MmpE's dual functions.
- (3) The study does not provide direct experimental validation of the MmpE deletion on lysosomal trafficking of the bacteria.
- (4) The role of MmpE as a mycobacterial effector would be more relevant using virulent mycobacterial strains such as H37Rv.

<https://doi.org/10.7554/eLife.108037.1.sa3>

Reviewer #2 (Public review):

Summary:

In this paper, the authors have characterized Rv2577 as a Fe³⁺/Zn²⁺-dependent metallophosphatase and a nucleomodulin protein. The authors have also identified His348 and Asn359 as critical residues for Fe³⁺ coordination. The authors show that the proteins encode for two nuclear localization signals. Using C-terminal Flag expression constructs, the authors have shown that the MmpE protein is secretory. The authors have prepared genetic deletion strains and show that MmpE is essential for intracellular survival of *M. bovis* BCG in THP-1 macrophages, RAW264.7 macrophages, and a mouse model of infection. The authors have also performed RNA-seq analysis to compare the transcriptional profiles of

macrophages infected with wild-type and MmpE mutant strains. The relative levels of ~ 175 transcripts were altered in MmpE mutant-infected macrophages and the majority of these were associated with various immune and inflammatory signalling pathways. Using these deletion strains, the authors proposed that MmpE inhibits inflammatory gene expression by binding to the promoter region of a vitamin D receptor. The authors also showed that MmpE arrests phagosome maturation by regulating the expression of several lysosome-associated genes such as TFEB, LAMP1, LAMP2, etc. These findings reveal a sophisticated mechanism by which a bacterial effector protein manipulates gene transcription and promotes intracellular survival.

Strength:

The authors have used a combination of cell biology, microbiology, and transcriptomics to elucidate the mechanisms by which Rv2577 contributes to intracellular survival.

Weakness:

The authors should thoroughly check the mice data and show individual replicate values in bar graphs.

<https://doi.org/10.7554/eLife.108037.1.sa2>

Reviewer #3 (Public review):

Summary:

In this manuscript titled "Mycobacterial Metallophosphatase MmpE Acts as a Nucleomodulin to Regulate Host Gene Expression and Promote Intracellular Survival", Chen et al describe biochemical characterisation, localisation and potential functions of the gene using a genetic approach in *M. bovis* BCG and perform macrophage and mice infections to understand the roles of this potentially secreted protein in the host cell nucleus. The findings demonstrate the role of a secreted phosphatase of *M. bovis* BCG in shaping the transcriptional profile of infected macrophages, potentially through nuclear localisation and direct binding to transcriptional start sites, thereby regulating the inflammatory response to infection.

Strengths:

The authors demonstrate using a transient transfection method that MmpE when expressed as a GFP-tagged protein in HEK293T cells, exhibits nuclear localisation. The authors identify two NLS motifs that together are required for nuclear localisation of the protein. A deletion of the gene in *M. bovis* BCG results in poorer survival compared to the wild-type parent strain, which is also killed by macrophages. Relative to the WT strain-infected macrophages, macrophages infected with the $\Delta mmpE$ strain exhibited differential gene expression. Overexpression of the gene in HEK293T led to occupancy of the transcription start site of several genes, including the Vitamin D Receptor. Expression of VDR in THP1 macrophages was lower in the case of $\Delta mmpE$ infection compared to WT infection. This data supports the utility of the overexpression system in identifying potential target loci of MmpE using the HEK293T transfection model. The authors also demonstrate that the protein is a phosphatase, and the phosphatase activity of the protein is partially required for bacterial survival but not for the regulation of the VDR gene expression.

Weaknesses:

(1) While the motifs can most certainly behave as NLSs, the overexpression of a mycobacterial protein in HEK293T cells can also result in artefacts of nuclear localisation. This is not unprecedented. Therefore, to prove that the protein is indeed secreted from BCG,

and is able to elicit transcriptional changes during infection, I recommend that the authors (i) establish that the protein is indeed secreted into the host cell nucleus, and (ii) the NLS mutation prevents its localisation to the nucleus without disrupting its secretion.

Demonstration that the protein is secreted: Supplementary Figure 3 - Immunoblotting should be performed for a cytosolic protein, also to rule out detection of proteins from lysis of dead cells. Also, for detecting proteins in the secreted fraction, it would be better to use Sauton's media without detergent, and grow the cultures without agitation or with gentle agitation. The method used by the authors is not a recommended protocol for obtaining the secreted fraction of mycobacteria.

Demonstration that the protein localises to the host cell nucleus upon infection: Perform an infection followed by immunofluorescence to demonstrate that the endogenous protein of BCG can translocate to the host cell nucleus. This should be done for an NLS1-2 mutant expressing cell also.

(2) In the RNA-seq analysis, the directionality of change of each of the reported pathways is not apparent in the way the data have been presented. For example, are genes in the cytokine-cytokine receptor interaction or TNF signalling pathway expressed more, or less in the *ΔmmpE* strain?

(3) Several of these pathways are affected as a result of infection, while others are not induced by BCG infection. For example, BCG infection does not, on its own, produce changes in IL1 β levels. As the authors did not compare the uninfected macrophages as a control, it is difficult to interpret whether *ΔmmpE* induced higher expression than the WT strain, or simply did not induce a gene while the WT strain suppressed expression of a gene. This is particularly important because the strain is attenuated. Does the attenuation have anything to do with the ability of the protein to induce lysosomal pathway genes? Does induction of this pathway lead to attenuation of the strain? Similarly, for pathways that seem to be downregulated in the *ΔmmpE* strain compared to the WT strain, these might have been induced upon infection with the WT strain but not sufficiently by the *ΔmmpE* strain due to its attenuation/ lower bacterial burden.

(4) CHIP-seq should be performed in THP1 macrophages, and not in HEK293T. Overexpression of a nuclear-localised protein in a non-relevant line is likely to lead to several transcriptional changes that do not inform us of the role of the gene as a transcriptional regulator during infection.

(5) I would not expect to see such large inflammatory reactions persisting 56 days post-infection with *M. bovis* BCG. Is this something peculiar for an intratracheal infection with 1x10⁷ bacilli? For images of animal tissue, the authors should provide images of the entire lung lobe with the zoomed-in image indicated as an inset.

(6) For the qRT-PCR based validation, infections should be performed with the *MmpE*-complemented strain in the same experiments as those for the WT and *ΔmmpE* strain so that they can be on the same graph, in the main manuscript file. Supplementary Figure 4 has three complementary strains. Again, the absence of the uninfected, WT, and *ΔmmpE* infected condition makes interpretation of these data very difficult.

(7) The abstract mentions that *MmpE* represses the PI3K-Akt-mTOR pathway, which arrests phagosome maturation. There is not enough data in this manuscript in support of this claim. Supplementary Figure 5 does provide qRT-PCR validation of genes of this pathway, but the data do not indicate that higher expression of these pathways, whether by VDR repression or otherwise, is driving the growth restriction of the *ΔmmpE* strain.

(8) The relevance of the NLS and the phosphatase activity is not completely clear in the CFU assays and in the gene expression data. Firstly, there needs to be immunoblot data provided

for the expression and secretion of the NLS-deficient and phosphatase mutants. Secondly, CFU data in Figure 3A, C, and E must consistently include both the WT and $\Delta mmpE$ strain.

<https://doi.org/10.7554/eLife.108037.1.sa1>

Author response:

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(1) The study addresses the role of nucleomodulins, an understudied aspect in mycobacterial infections.

(2) The authors employ a combination of biochemical and computational analyses along with in vitro and in vivo validations to characterize the role of MmpE.

Weaknesses:

(1) While the study establishes that the phosphatase activity of MmpE operates independently of its NLS, there is a clear gap in understanding how this phosphatase activity supports mycobacterial infection. The investigation lacks experimental data on specific substrates of MmpE or pathways influenced by this virulence factor.

We thank the reviewer for this insightful comment and agree that identification of the substrate of MmpE is important to fully understand its role in mycobacterial infection.

MmpE is a putative purple acid phosphatase (PAP) and a member of the metallophosphoesterase (MPE) superfamily. Enzymes in this family are known for their catalytic promiscuity and broad substrate specificity, acting on phosphomonoesters, phosphodiesteres, and phosphotriesters (Matange et al., Biochem J., 2015). In bacteria, several characterized MPEs have been shown to hydrolyze substrates such as cyclic nucleotides (e.g., cAMP) (Keppetipola et al., J Biol Chem, 2008; Shenoy et al., J Mol Biol, 2007), nucleotide derivatives (e.g., AMP, UDP-glucose) (Innokentev et al., mBio, 2025), and pyrophosphate-containing compounds (e.g., Ap4A, UDP-DAGn) (Matange et al., Biochem J., 2015). Although the binding motif of MmpE has been identified, determining its physiological substrates remains challenging due to the low abundance and instability of potential metabolites, as well as the limited sensitivity and coverage of current metabolomic technologies in mycobacteria.

(2) The study does not explore whether the phosphatase activity of MmpE is dependent on the NLS within macrophages, which would provide critical insights into its biological relevance in host cells. Conducting experiments with double knockout/mutant strains and comparing their intracellular survival with single mutants could elucidate these dependencies and further validate the significance of MmpE's dual functions.

We thank the reviewer for the comment. In our study, we demonstrate that both the nuclear localization and phosphatase activity of MmpE are required for full virulence (Figure 3D–E). Importantly, deletion of the NLS motifs did not impair MmpE’s phosphatase activity in vitro (Figure 2F), indicating that its enzymatic function is structurally independent of its nuclear localization. These findings suggest that MmpE functions as a bifunctional protein, with distinct and non-overlapping roles for its nuclear trafficking and phosphatase activity. We have expanded on this point in the Discussion section “MmpE Functions as a Bifunctional Protein with Nuclear Localization and Phosphatase Activity”.

(3) *The study does not provide direct experimental validation of the MmpE deletion on lysosomal trafficking of the bacteria.*

We thank the reviewer for the comment. The role of Rv2577/MmpE in phagosome maturation has been demonstrated in *M. tuberculosis*, where its deletion increases colocalization with lysosomal markers such as LAMP-2 and LAMP-3 (Forrellad et al., Front Microbiol, 2020). In our study, we found that mmpE deletion in *M. bovis* BCG led to upregulation of lysosomal genes, including TFEB, LAMP1, LAMP2, and v-ATPase subunits, compared to the wild-type strain. These results suggest that MmpE may regulate lysosomal trafficking by interfering with phagosome–lysosome fusion.

To further validate MmpE’s role in phagosome maturation, we will perform fluorescence colocalization assays in THP-1 macrophages infected with BCG/wt, Δ mmpE, complemented, and NLS-mutant strains. Co-staining with LAMP1 and LysoTracker will allow us to assess whether the Δ mmpE mutant is more efficiently trafficked to lysosomes.

(4) *The role of MmpE as a mycobacterial effector would be more relevant using virulent mycobacterial strains such as H37Rv.*

We thank the reviewer for the comment. Previously, the role of Rv2577/MmpE as a virulence factor has been demonstrated in *M. tuberculosis* CDC 1551, where its deletion significantly reduced bacterial replication in mouse lungs at 30 days post-infection (Forrellad et al., Front Microbiol, 2020). However, that study did not explore the underlying mechanism of MmpE function. In our work, we found that MmpE enhances *M. bovis* BCG survival in both macrophages (THP-1 and RAW264.7) and mice (Figure 2A–B, Figure 6A), consistent with its proposed role in virulence. To investigate the molecular mechanism by which MmpE promotes intracellular survival, we used *M. bovis* BCG as a biosafe surrogate and this model is widely accepted for studying mycobacterial pathogenesis (Wang et al., Nat Immunol, 2025; Wang et al., Nat Commun, 2017; Péan et al., Nat Commun, 2017).

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*In this paper, the authors have characterized Rv2577 as a Fe³⁺/Zn²⁺ -dependent metallophosphatase and a nucleomodulin protein. The authors have also identified His348 and Asn359 as critical residues for Fe³⁺ coordination. The authors show that the proteins encode for two nuclear localization signals. Using C-terminal Flag expression constructs, the authors have shown that the MmpE protein is secretory. The authors have prepared genetic deletion strains and show that MmpE is essential for intracellular survival of *M. bovis* BCG in THP-1 macrophages, RAW264.7 macrophages, and a mouse model of infection. The authors have also performed RNA-seq analysis to compare the transcriptional profiles of macrophages infected with wild-type and MmpE mutant strains. The relative levels of ~ 175 transcripts were altered in MmpE mutant-infected macrophages and the majority of these were associated with various immune and*

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Strength:

The authors have used a combination of cell biology, microbiology, and transcriptomics to elucidate the mechanisms by which Rv2577 contributes to intracellular survival.

Weakness:

The authors should thoroughly check the mice data and show individual replicate values in bar graphs.

We kindly appreciate the reviewer for the advice. We will update the relevant mice data in the revised manuscript.

Reviewer #3 (Public review):

Summary:

*In this manuscript titled "Mycobacterial Metallophosphatase MmpE Acts as a Nucleomodulin to Regulate Host Gene Expression and Promote Intracellular Survival", Chen et al describe biochemical characterisation, localisation and potential functions of the gene using a genetic approach in *M. bovis* BCG and perform macrophage and mice infections to understand the roles of this potentially secreted protein in the host cell nucleus. The findings demonstrate the role of a secreted phosphatase of *M. bovis* BCG in shaping the transcriptional profile of infected macrophages, potentially through nuclear localisation and direct binding to transcriptional start sites, thereby regulating the inflammatory response to infection.*

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authors (i) establish that the protein is indeed secreted into the host cell nucleus, and (ii) the NLS mutation prevents its localisation to the nucleus without disrupting its secretion.

We kindly appreciate the reviewer for the advice and will include the relevant experiments in the revised manuscript. The localization of WT MmpE and the NLS mutated MmpE will be tested in the BCG infected macrophages.

Demonstration that the protein is secreted: Supplementary Figure 3 - Immunoblotting should be performed for a cytosolic protein, also to rule out detection of proteins from lysis of dead cells. Also, for detecting proteins in the secreted fraction, it would be better to use Sauton's media without detergent, and grow the cultures without agitation or with gentle agitation. The method used by the authors is not a recommended protocol for obtaining the secreted fraction of mycobacteria.

We agree with the reviewer and we will further validate the secretion of MmpE using the tested protocol.

Demonstration that the protein localises to the host cell nucleus upon infection: Perform an infection followed by immunofluorescence to demonstrate that the endogenous protein of BCG can translocate to the host cell nucleus. This should be done for an NLS1-2 mutant expressing cell also.

We will add this experiment in the revised manuscript.

(2) In the RNA-seq analysis, the directionality of change of each of the reported pathways is not apparent in the way the data have been presented. For example, are genes in the cytokine-cytokine receptor interaction or TNF signalling pathway expressed more, or less in the Δ mmpE strain?

We thank the reviewer for pointing this out and fully agree that conventional KEGG pathway enrichment diagrams do not convey the directionality of individual gene expression changes within each pathway. While KEGG enrichment analysis identifies pathways that are statistically overrepresented among differentially expressed genes, it does not indicate whether individual genes within those pathways are upregulated or downregulated.

To address this, we re-analyzed the expression trends of DEGs within each significantly enriched KEGG pathway. The results show that key immune-related pathways, including cytokine-cytokine receptor interaction, TNF signaling, NF- κ B signaling, and chemokine signaling, are collectively upregulated in THP-1 macrophages infected with Δ mmpE strain compared to those infected with the wild-type BCG strain. The full list of DEGs will be provided in the supplementary materials. The complete RNA-seq dataset has been deposited in the GEO database, and the accession number will be included in the revised manuscript.

(3) Several of these pathways are affected as a result of infection, while others are not induced by BCG infection. For example, BCG infection does not, on its own, produce changes in IL1 β levels. As the author s did not compare the uninfected macrophages as a control, it is difficult to interpret whether Δ mmpE induced higher expression than the WT strain, or simply did not induce a gene while the WT strain suppressed expression of a gene. This is particularly important because the strain is attenuated. Does the attenuation have anything to do with the ability of the protein to induce lysosomal pathway genes? Does induction of this pathway lead to attenuation of the strain? Similarly, for pathways that seem to be downregulated in the Δ mmpE strain compared to the WT strain, these might have been induced upon infection with the WT strain but not sufficiently by the Δ mmpE strain due to its attenuation/ lower bacterial burden.

We thank the reviewer for the comment. We will update qRT-PCR data with the uninfected macrophages as a control in the revised manuscript.

Wild-type *Mycobacterium bovis* BCG strain still has the function of inhibiting phagosome maturation (Branzk et al., Nat Immunol, 2014; Weng et al., Nat Commun, 2022). Forrellad et al. previously identified Rv2577/MmpE as a virulence factor in *M. tuberculosis* and disruption of the MmpE gene impairs the ability of *M. tuberculosis* to arrest phagosome maturation (Forrellad et al., Front Microbiol, 2020). In our study, transcriptomic and qRT-PCR data (Figures 4C and G, S4C) show that deletion of mmpE in *M. bovis* BCG leads to upregulation of lysosomal biogenesis and acidification genes, including TFEB, LAMP1, and vATPase. To further validate MmpE's role in phagosome maturation, we will perform fluorescence colocalization assays in THP-1 macrophages infected with BCG/wt, Δ mmpE, complemented, and NLS-mutant strains. Co-staining with LAMP1 and LysoTracker will assess whether the Δ mmpE mutant is more efficiently trafficked to lysosomes.

Furthermore, CFU assays demonstrated that the Δ mmpE strain exhibits markedly reduced bacterial survival in both human THP-1 and murine RAW264.7 macrophages, as well as in mice, compared to the wild-type strain (Figures 4A and C, 6A). These findings suggest that the loss of MmpE compromises bacterial survival, likely due to enhanced lysosomal trafficking and acidification. This supports previous studies showing that increased lysosomal activity promotes mycobacterial clearance (Gutierrez et al., Cell, 2004; Pilli et al., Immunity, 2012).

(4) ChIP-seq should be performed in THP1 macrophages, and not in HEK293T. Overexpression of a nuclear-localised protein in a non-relevant line is likely to lead to several transcriptional changes that do not inform us of the role of the gene as a transcriptional regulator during infection.

We thank the reviewer for the comment. We performed ChIP-seq in HEK293T cells is based on the fact that this cell line is widely used in ChIP-based assays due to its high transfection efficiency, robust nuclear protein expression, and well-annotated genome (Lampe et al., Nat Biotechnol, 2024; Marasco et al., Cell, 2022). These features make HEK293T an ideal system for the initial identification of genome wide chromatin binding profiles of novel nuclear effectors such as MmpE.

Furthermore, we validated the major observations in THP-1 macrophages, including (i) RNAseq of THP-1 cells infected with either WT BCG or Δ mmpE strains revealed significant transcriptional changes in immune and lysosomal pathways (Figure 4A); (ii) Integrated analysis of CUT&Tag and RNA-seq data identified 298 genes in infected THP-1 cells that exhibited both MmpE binding and corresponding expression changes. Among these, VDR was validated as a direct transcriptional target of MmpE using EMSA and ChIP-PCR (Figures 5E-J, S5D-F). Notably, the signaling pathways associated with MmpE-bound genes, including PI3K-Akt-mTOR signaling and lysosomal function, substantially overlap with those transcriptionally modulated in infected THP-1 macrophages (Figures 4B-G, S4B-C, S5C-D), further supporting the biological relevance of the ChIP-seq data obtained from HEK293T cells.

(5) I would not expect to see such large inflammatory reactions persisting 56 days postinfection with M. bovis BCG. Is this something peculiar for an intratracheal infection with 1x10⁷ bacilli? For images of animal tissue, the authors should provide images of the entire lung lobe with the zoomed-in image indicated as an inset.

We thank the reviewer for the comment. The lung inflammation peaked at days 21–28 and had clearly subsided by day 56 across all groups (Figure 6B), consistent with the expected resolution of immune responses to an attenuated strain like *M. bovis* BCG. This temporal pattern is in line with previous studies using intravenous or intratracheal BCG vaccination in

mice and macaques, which also demonstrated robust early immune activation followed by resolution over time (Smith et al., Nat Microbiol, 2025; Darrah et al., Nature, 2020).

In this study, the infectious dose (1×10^7 CFU intratracheally) was selected based on previous studies in which intratracheal delivery of 1×10^7 CFU produced consistent and measurable lung immune responses and pathology without causing overt illness or mortality (Xu et al., Sci Rep, 2017; Niroula et al., Sci Rep, 2025). We will provide whole-lung lobe images with zoomed-in insets in the revised manuscript.

(6) For the qRT-PCR based validation, infections should be performed with the MmpE complemented strain in the same experiments as those for the WT and Δ mmpE strain so that they can be on the same graph, in the main manuscript file. Supplementary Figure 4 has three complementary strains. Again, the absence of the uninfected, WT, and Δ mmpE infected condition makes interpretation of these data very difficult.

We thank the reviewer for the comment. As suggested, we will conduct the qRT-PCR experiment including the uninfected, WT, Δ mmpE, Comp-MmpE, and the three complementary strains infecting THP-1 cells. The updated data will be provided in the revised manuscript.

(7) The abstract mentions that MmpE represses the PI3K-Akt-mTOR pathway, which arrests phagosome maturation. There is not enough data in this manuscript in support of this claim. Supplementary Figure 5 does provide qRT-PCR validation of genes of this pathway, but the data do not indicate that higher expression of these pathways, whether by VDR repression or otherwise, is driving the growth restriction of the Δ mmpE strain.

We thank the reviewer for the comment. The role of MmpE in phagosome maturation was previously characterized. Disruption of mmpE impairs the ability of *M. tuberculosis* to arrest lysosomal trafficking (Forrellad et al., Front Microbiol, 2020). In this study, we further found that MmpE suppresses the expression of key lysosomal genes, including TFEB, LAMP1, LAMP2, and ATPase subunits (Figure 4G), suggesting MmpE is involved in arresting phagosome maturation. As noted, the genes in the PI3K-Akt-mTOR pathway are upregulated in Δ mmpE-infected macrophages (Figure S5C).

To functionally validate this, we will conduct two complementary experimental approaches:

(i) Immunofluorescence assays: We will assess phagosome maturation and lysosomal fusion in THP-1 cells infected with BCG/wt, Δ mmpE, Comp-MmpE, and NLS mutant strains. Colocalization of intracellular bacteria with LAMP1 and LysoTracker will be quantified to determine whether the Δ mmpE strain is more efficiently trafficked to lysosomes.

(ii) CFU assays: We will perform CFU assays in THP-1 cells infected with BCG/wt or Δ mmpE in the presence or absence of PI3K-Akt-mTOR pathway inhibitors (e.g., Dactolisib), to assess whether activation of this pathway contributes to the intracellular growth restriction observed in the Δ mmpE strain.

(8) The relevance of the NLS and the phosphatase activity is not completely clear in the CFU assays and in the gene expression data. Firstly, there needs to be immunoblot data provided for the expression and secretion of the NLS-deficient and phosphatase mutants. Secondly, CFU data in Figure 3A, C, and E must consistently include both the WT and Δ mmpE strain.

We thank the reviewer for the comment. We will provide immunoblot data for the expression and secretion of the NLS-deficient and phosphatase mutants. Additionally, we will revise

Figure 3A, 3C, and 3E to consistently include both the WT and Δ mmpE strains in the CFU assays.

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